

Relations

Social Networks Theories and Methods

James Hollway

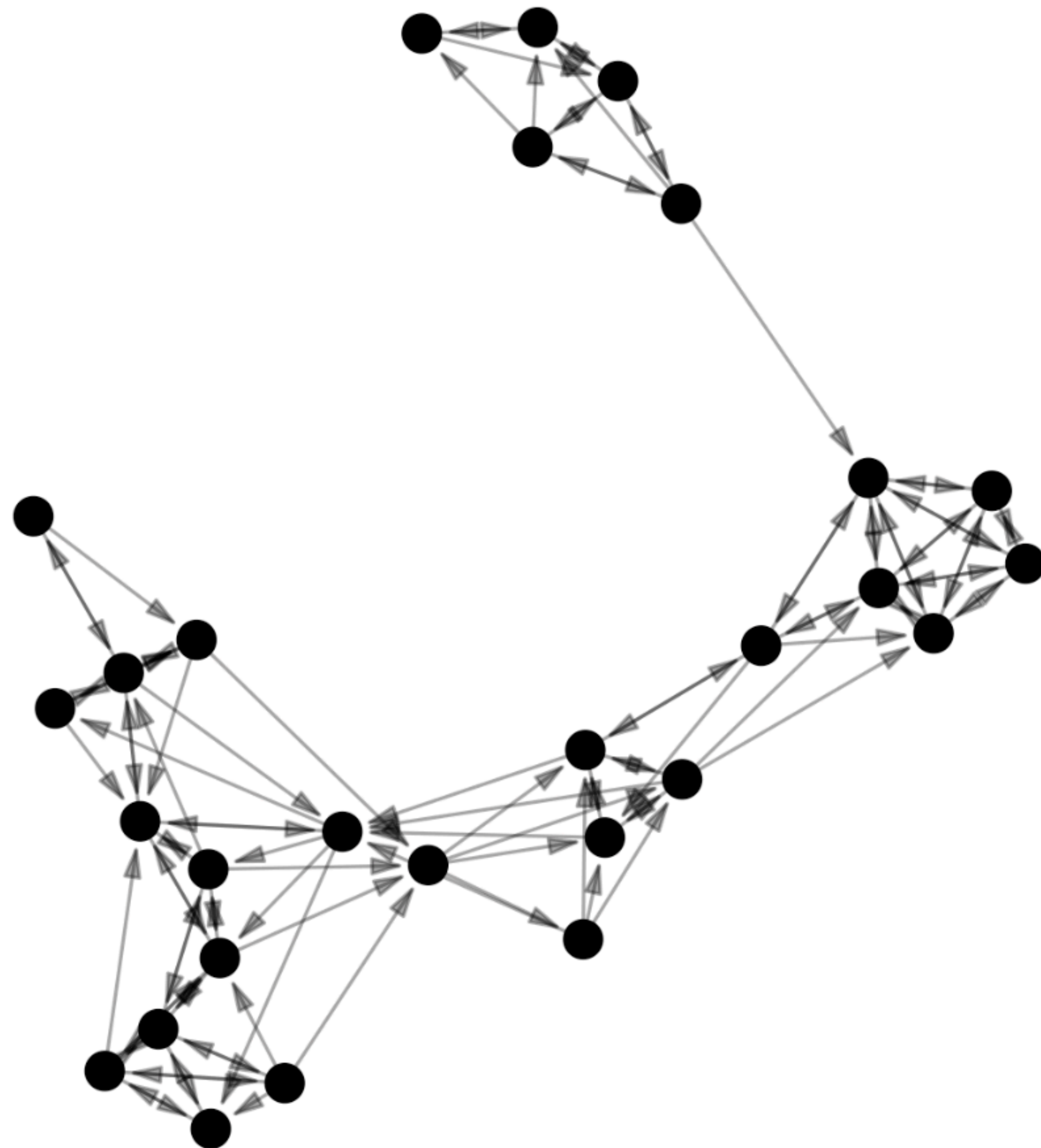
Housekeeping!

- Slack
- AI
- Midterm
- Next week

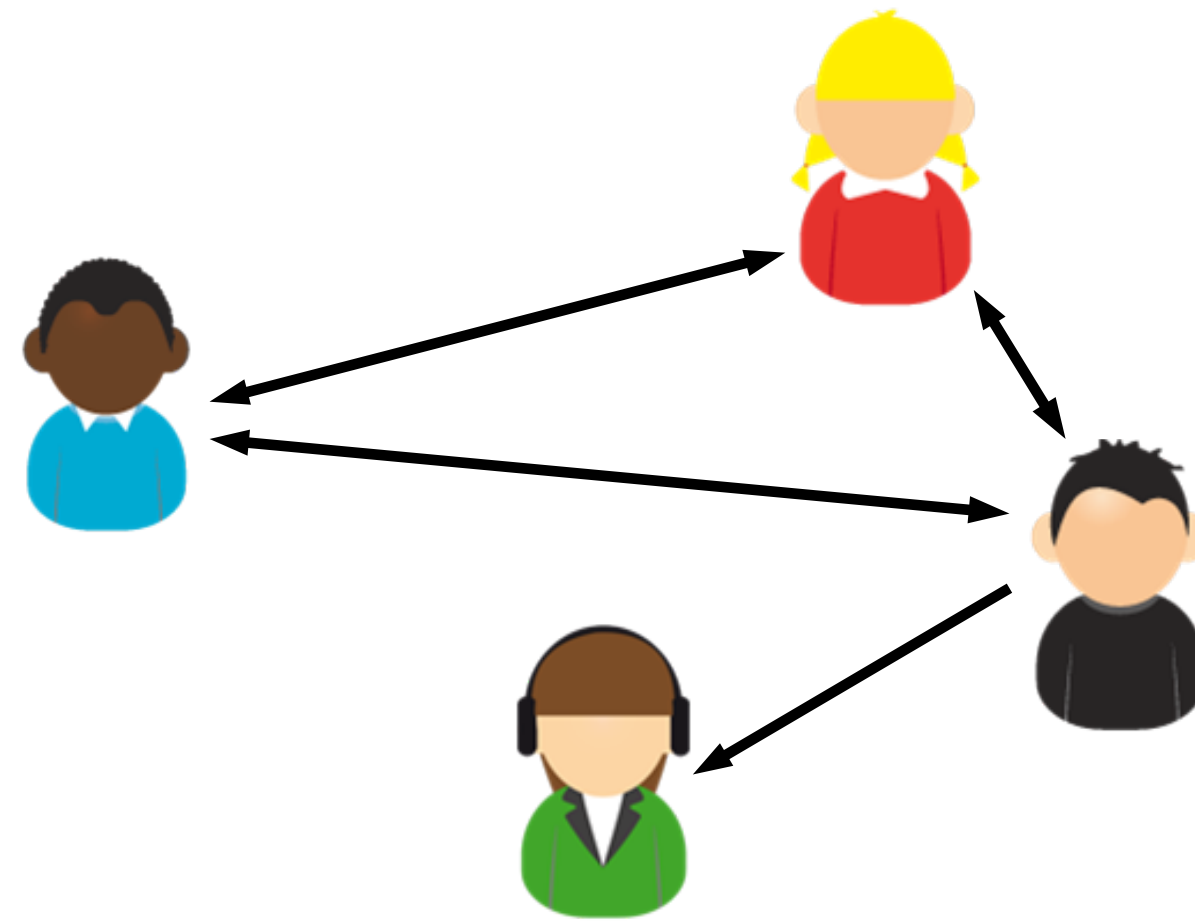
Memory check



- How would you describe this network?



Graph

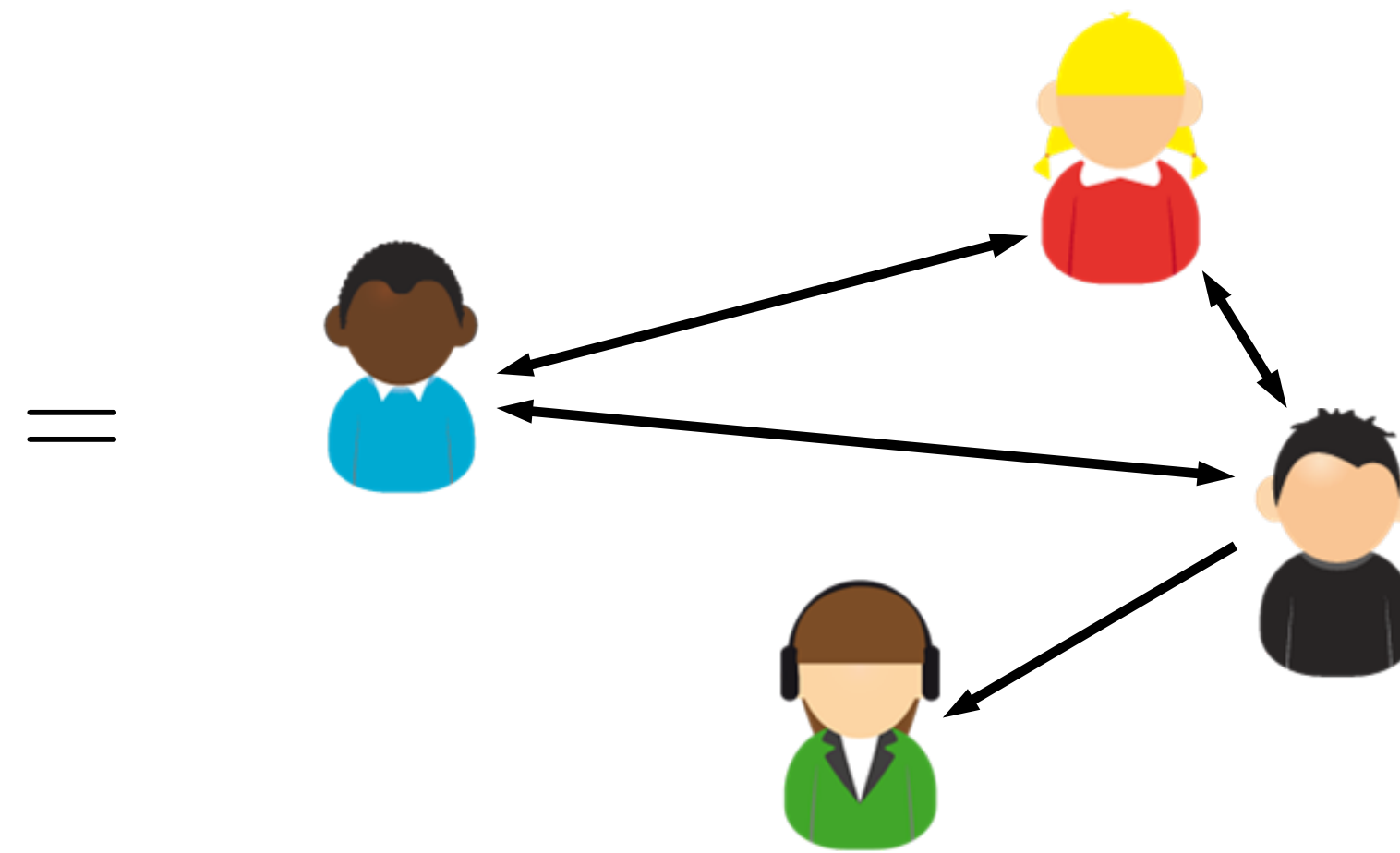


- + graphs are pretty
- + often surprisingly natural to interpret
- can quickly become complex and details difficult to discern (the ‘hairball’)
- different eyes may see different things

See `autograph::graphr()`, next week

Edgelist $\leftrightarrow$ Graph

consist of 2
numbered or labelled
columns (if directed
then ordered)

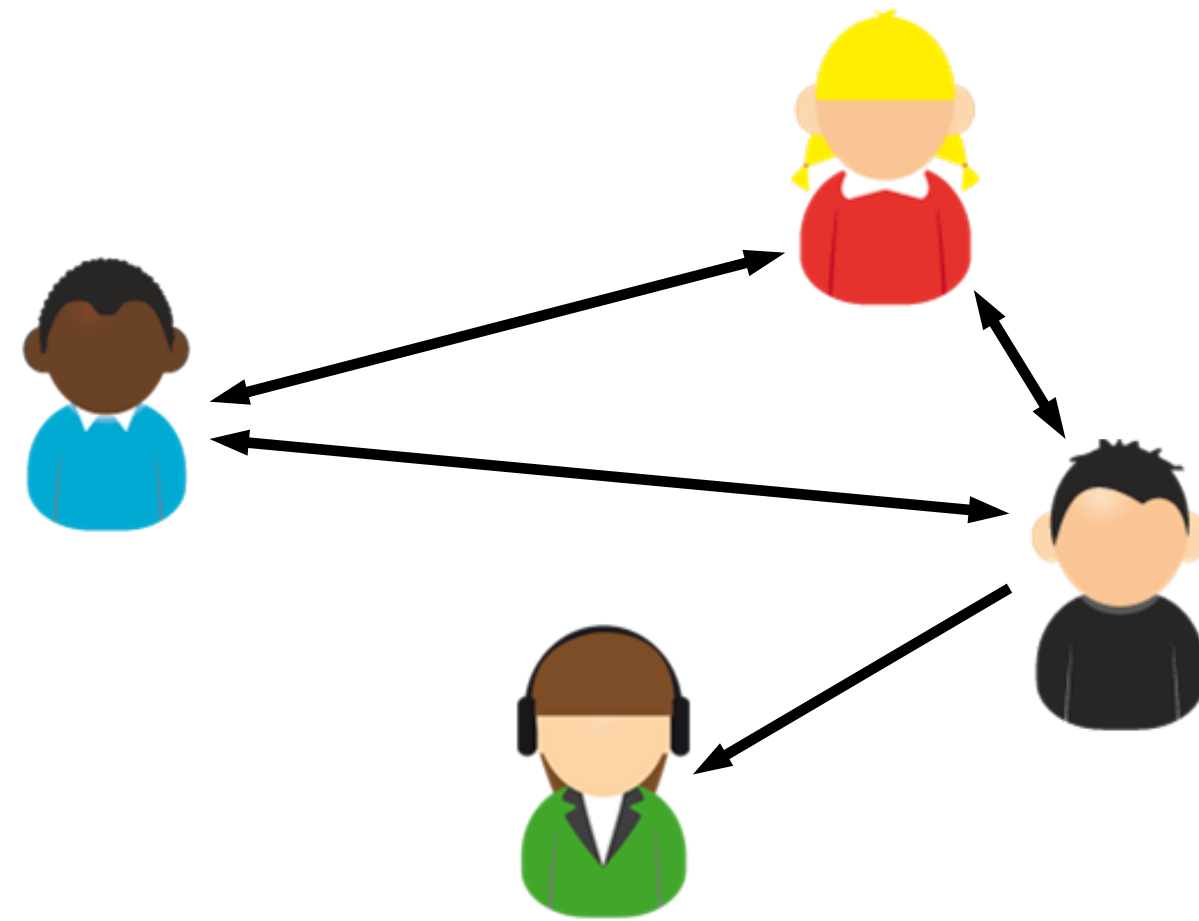


- + easy to create and edit in Excel/Numbers, and memory efficient
 - + can add edge information (time, weight, etc) as additional columns
 - cannot deal with isolates/nodal attributes without additional data objects
 - more complicated statistics than degree difficult to calculate
- See `as_edgelist()`

Edgelist $\leftrightarrow$ Graph $\leftrightarrow$ Matrix



=



=

$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

rows and columns correspond to (ordered)
senders and recipients

0 indicates absence of a tie (diagonal 0s in
simple network)

1 (or more) indicates a tie from sender to
recipient

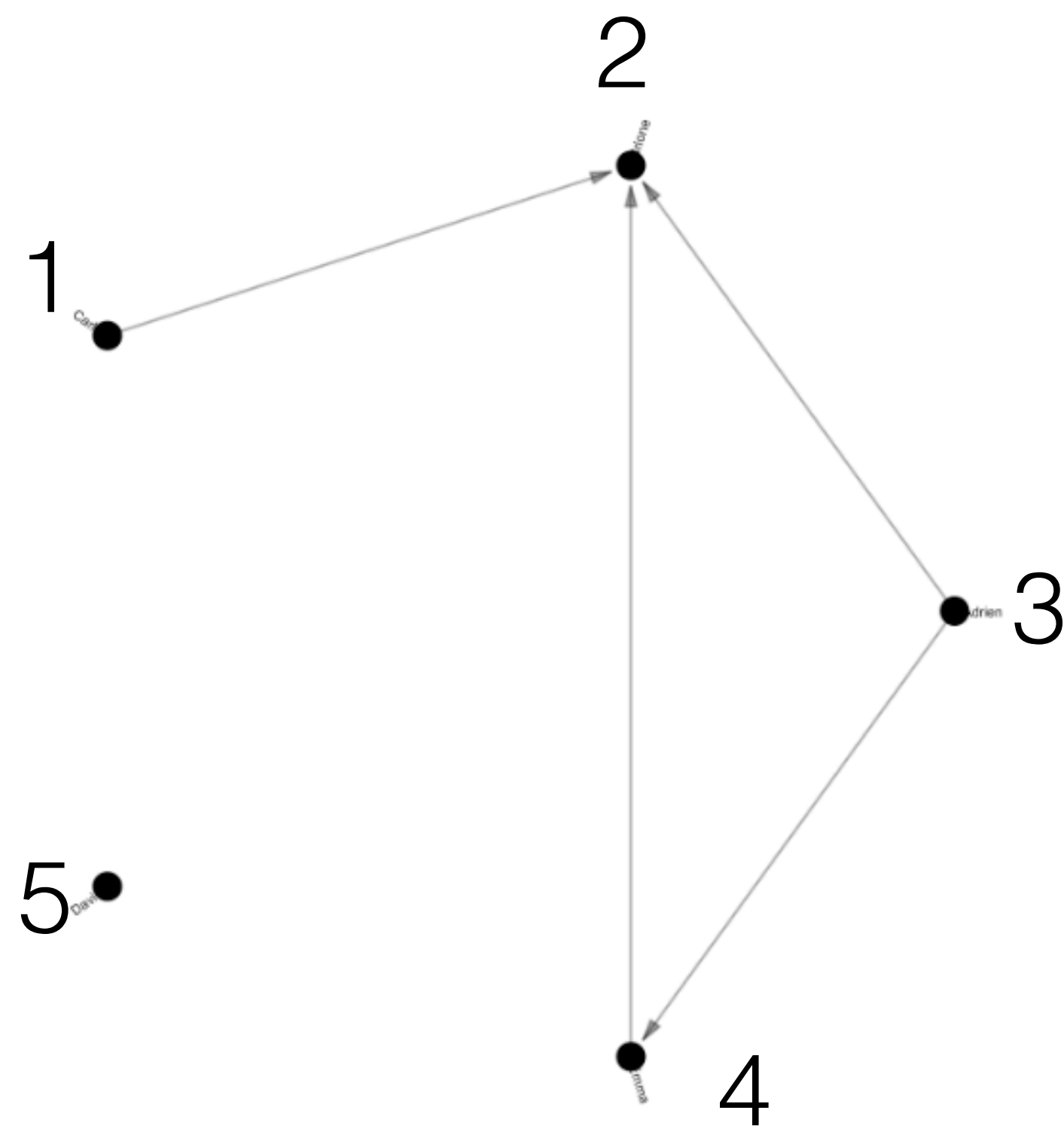
- memory inefficient for sparse networks
- somewhat incomprehensible
- + encodes all relational information (ties, weights, isolates, etc)
- + flexible, *quick* analysis via matrix algebra

See `as_matrix()`

“We dream of graphs,
but calculate in matrices”

Barry Wellman
(1942–2024)

Let's try a few

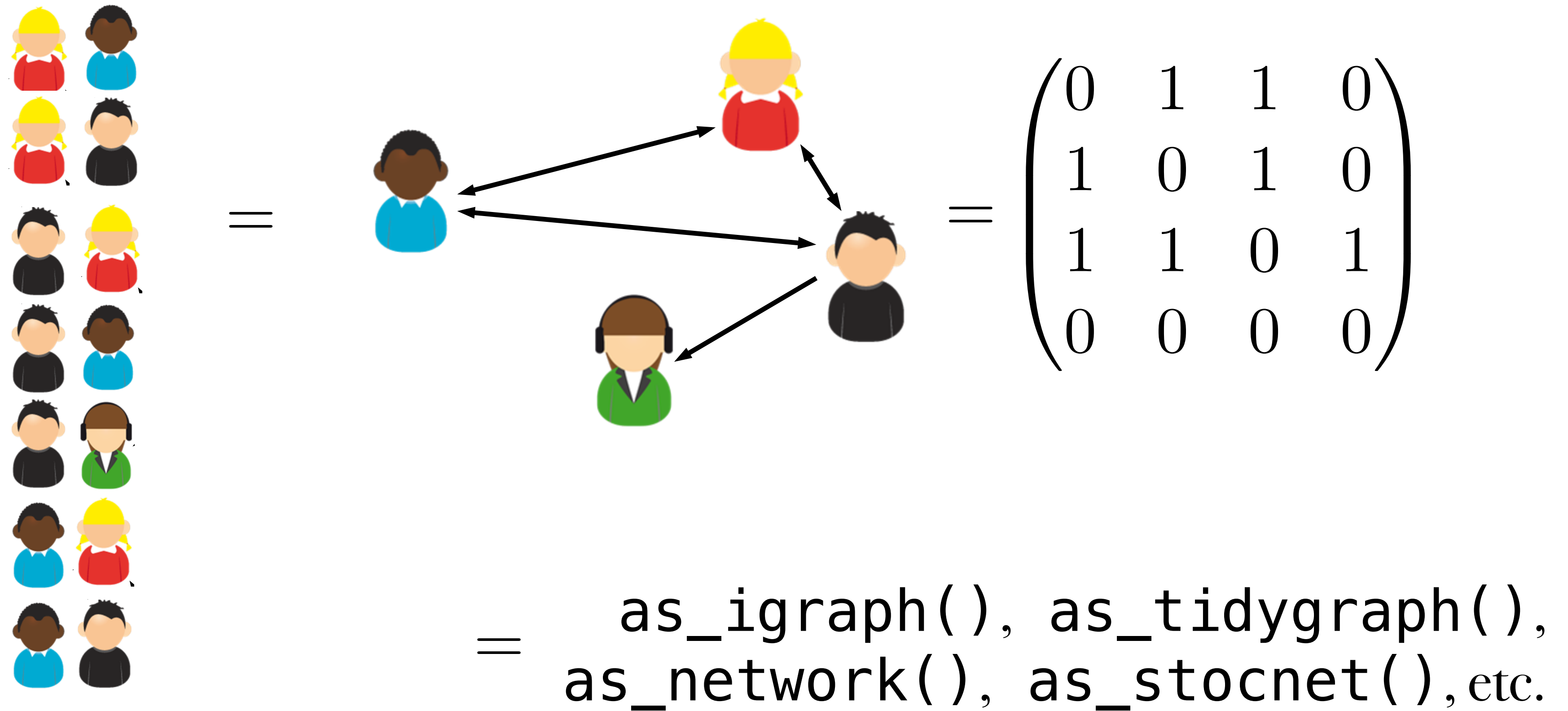


##		[,1]	[,2]	[,3]	[,4]
##	[1,]	0	0	1	0
##	[2,]	1	0	0	0
##	[3,]	1	0	1	1
##	[4,]	0	1	1	1

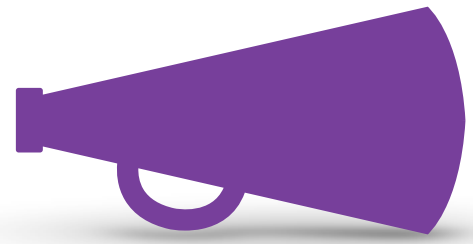
##	from	to
##	1	2
##	2	1
##	3	2
##	4	2
##	5	3
##	6	4

Turn into matrix, edgelist, and graph respectively

Edgelist $\leftrightarrow$ Graph $\leftrightarrow$ Matrix $\leftrightarrow$ Network



Where do ties come from? Some sources



Self-reports

- Surveys (e.g. Hogan et al 2016)
 - using e.g. [Network Canvas](#)
 - name generators vs rosters
- Interviews (e.g. Bellotti 2014)



Constructions

- Web-scraping
 - Chrome or Firefox extensions
 - Rcrawler, rvest and vosonSML packages
 - Python (e.g. to access Twitter API)
- Manual/automatic text coding
 - Considerable literature-based datasets
 - Can extract relationships (e.g. similarities) in or across documents



Observation

- Participant observation (e.g. Wyatt et al 2011)
- RFID badges (e.g. Elmer et al 2018)

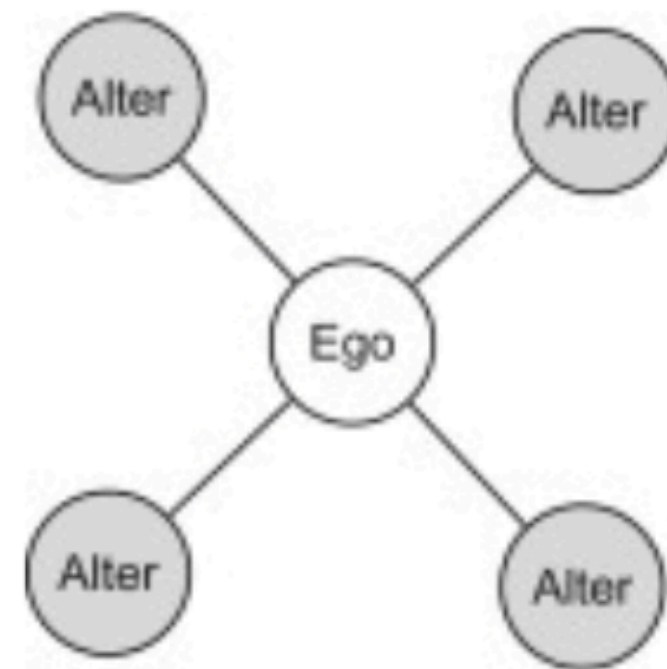


Archives

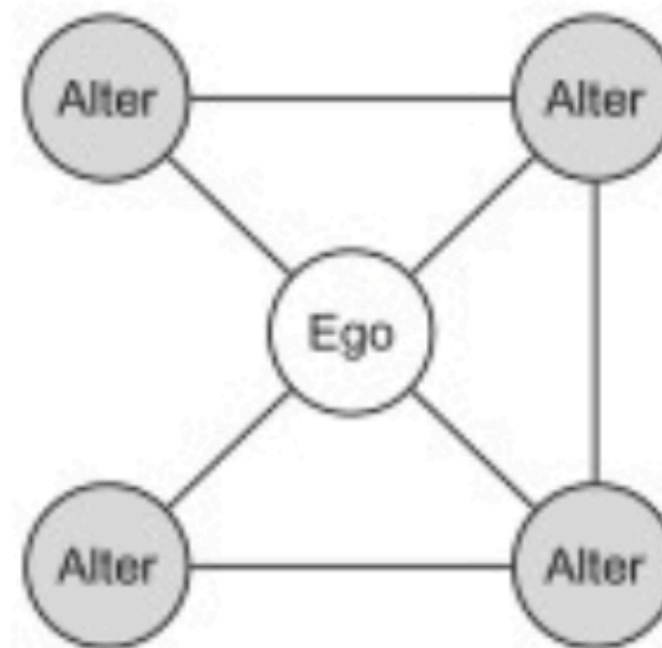
- Individual
 - Diary research (e.g. Fu 2008)
 - Historical records (e.g. Padgett & Ansell 1993)
 - Social media (e.g. Golder et al 2007)
 - Communication logs (e.g. Goldberg et al 2016)
- Organizational
 - Publication or patent records (e.g. Lazega et al 2008, Goetze 2010)
 - Agreement data between countries (e.g. Hollway and Koskinen 2016)

Ego networks

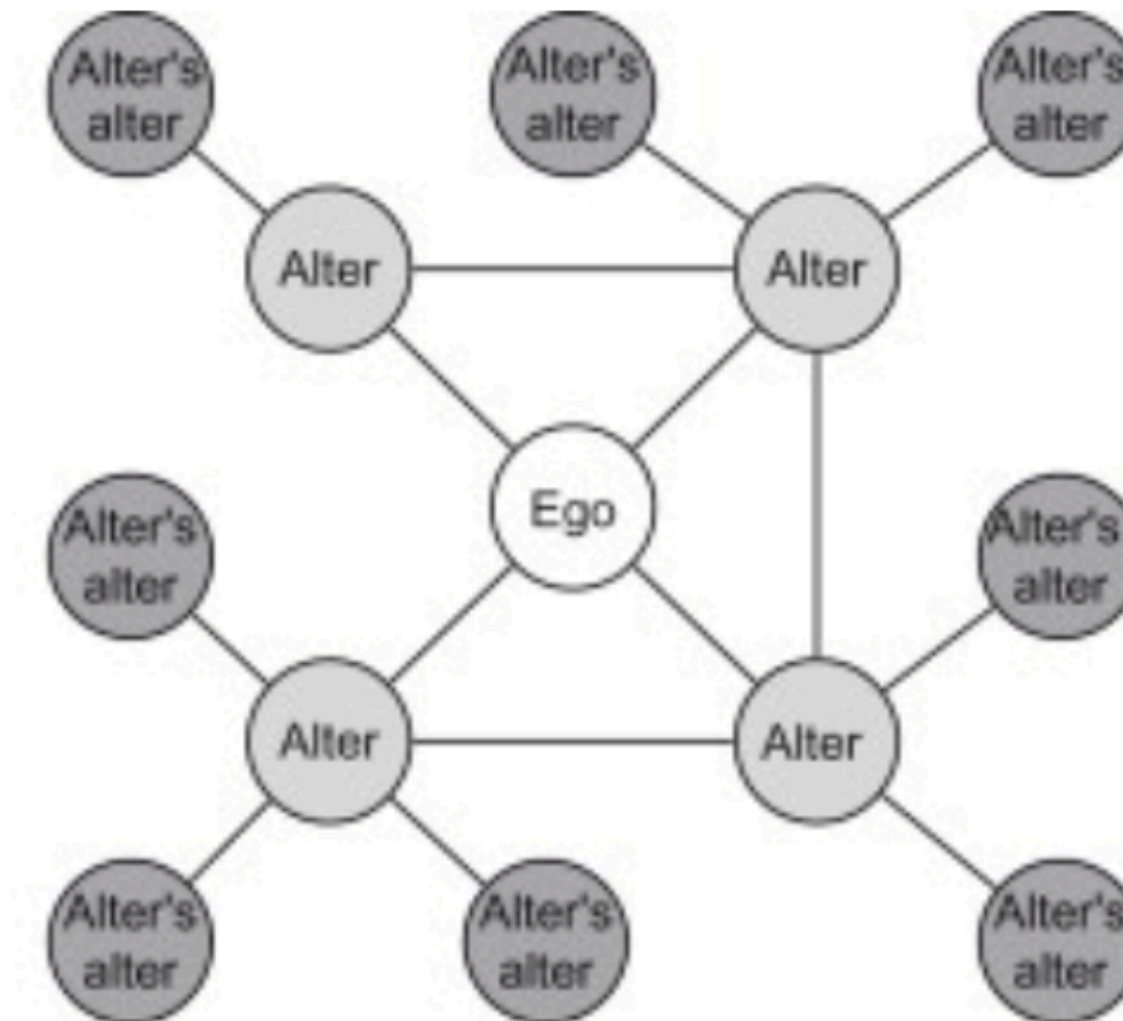
1.0
first-order star



1.5
first-order zone

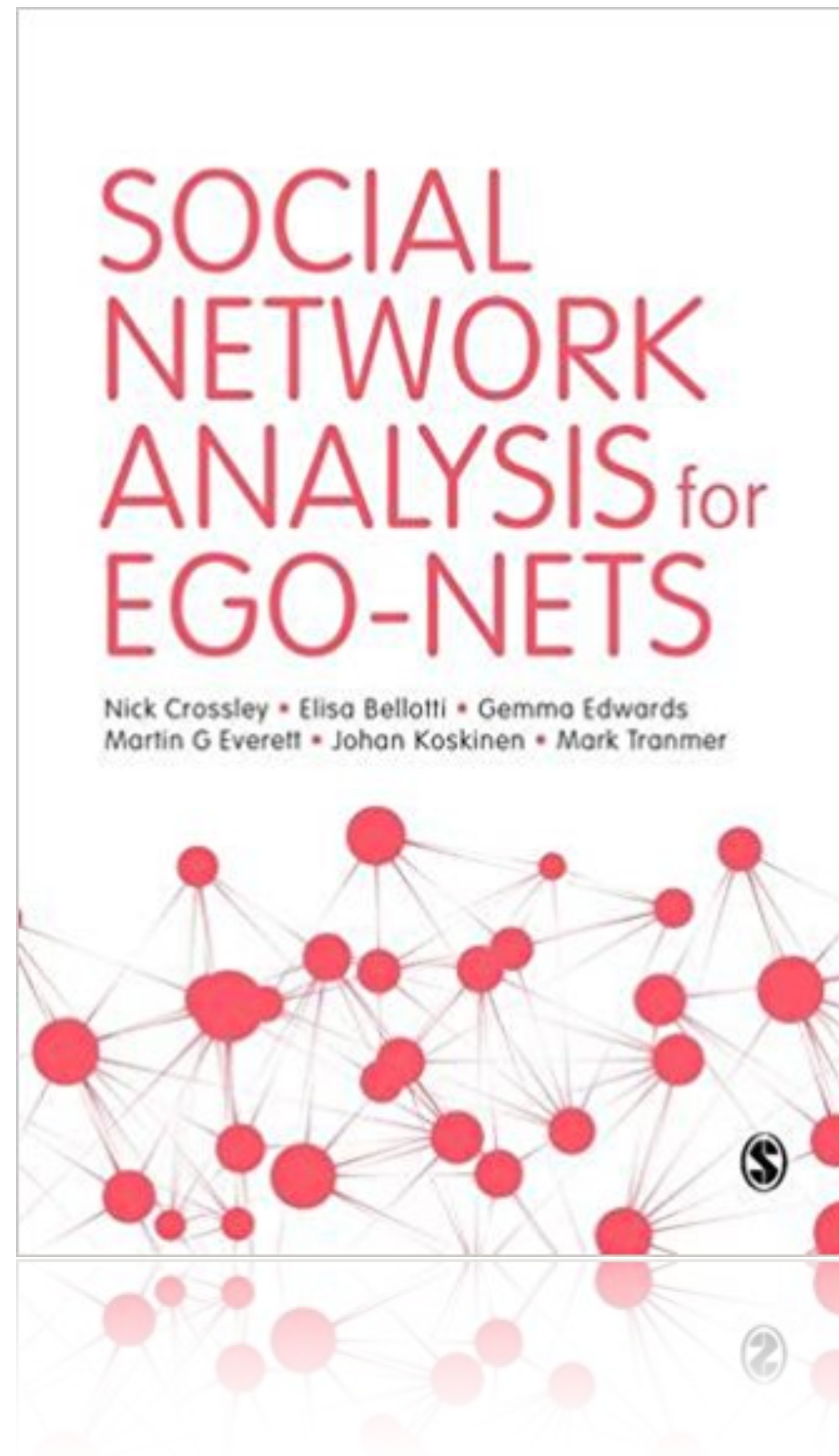


2.0
second-order star

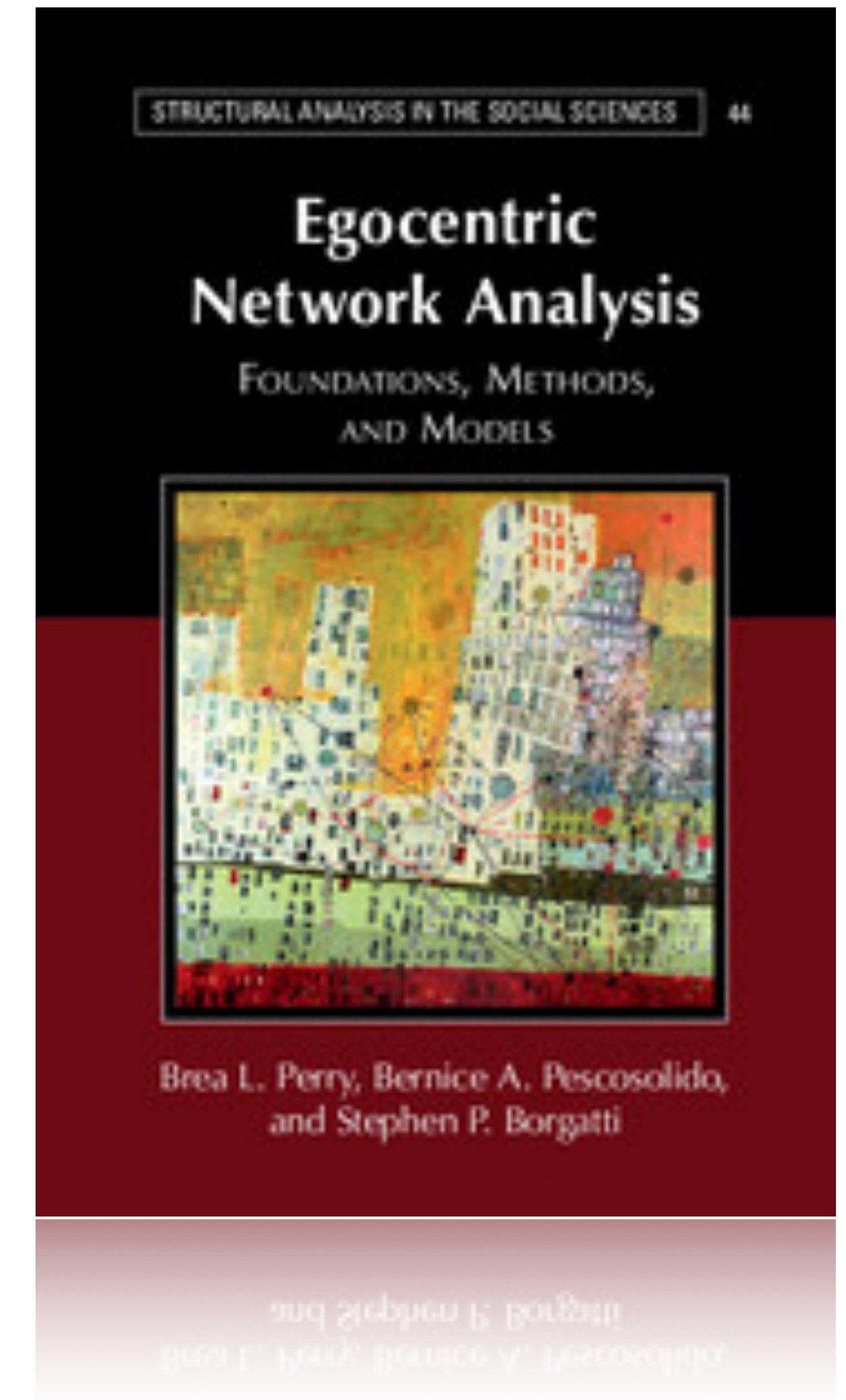


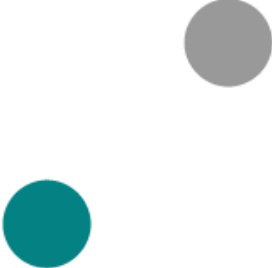
- Can analyse larger networks (sampling)
- Compatible with traditional methods (and designs)
- Can study intersecting social circles (through focal nodes)

Collection strategies

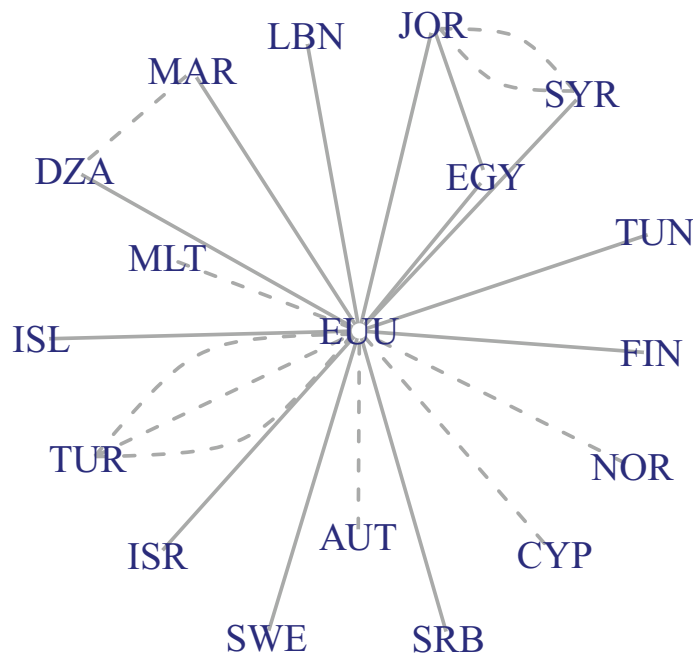


- Name *generator*: Elicits respondents to report names of all people with whom they discussed important matters in last six months
- Name *interpreter*: Collects information on characteristics of first five people named (Burt 1984, Marsden 1987)
- Name *interrelater*: Moves beyond ego-alter dyads to capture the perceived structure of the ego's network



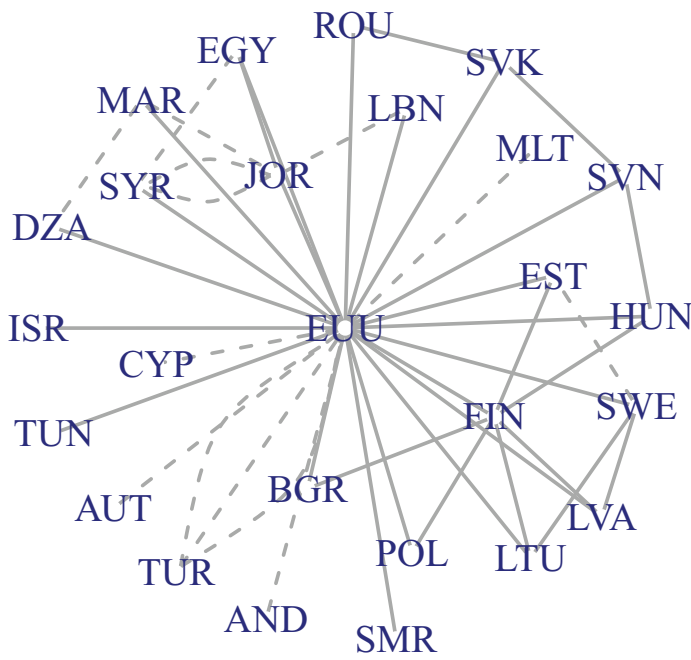
ONE-PERSON	NUCLEAR	NUCLEAR	NUCLEAR	NUCLEAR	COMPOSITE	NUCLEAR	NUCLEAR	NUCLEAR	COMPOSITE
One-person household 	Married couple 	Married couple, kid 	Married couple, 2 kids 	Parent, 1 kid 	Householder, friend or partner 	Married couple, 3 kids 	Parent, 2 kids 	Married couple, 4 kids 	Parent, kid, friend or partner 
NUCLEAR	COMPOSITE	EXTENDED	COMPOSITE	EXTENDED	EXTENDED	EXTENDED	EXTENDED	EXTENDED	EXTENDED
Parent, 3 kids 	Householder, non-relative 	Householder, sibling 	Parent, 2 kids, friend or partner 	Householder, parent 	Parent, kid, grandkid 	Householder, grandkid 	Married couple, grandkid 	Married couple, kid, grandkid 	Householder, relative 
COMPOSITE	COMPOSITE	NUCLEAR	NUCLEAR	EXTENDED	COMPOSITE	COMPOSITE	EXTENDED	EXTENDED	EXTENDED
Group of 3 friends 	Parent, 3 kids, friend or partner 	Married couple, 5 kids 	Parent, 4 kids 	Married couple, grandparent 	Householder, friend, non-rel. 	Parent, kid, non-relative 	Parent, kid, 2 grandkids 	Married couple, relative 	Married couple, 2 kids, grandkid 
EXTENDED	EXTENDED	EXTENDED	COMPOSITE	COMPOSITE	EXTENDED	COMPOSITE	EXTENDED	EXTENDED	COMPOSITE
Married couple, 1 kid, 2 grandkids 	Married couple, 2 kids, relative 	Householder, kid, parent 	Non-related group of 3 	Married couple, non-relative 	Married couple, kid, parent 	Group of 4 friends 	Parent, 2 kids, grandkid 	Parent, kid, grandparent 	Married couple, kid, non-relative 
EXTENDED	EXTENDED	EXTENDED	EXTENDED	NUCLEAR	EXTENDED	EXTENDED	NUCLEAR	EXTENDED	COMPOSITE
Married couple, relative 	Married couple, sibling 	Married couple, 2 kids, parent-in-law 	Parent, kid, sibling 	Parent, kid, stepkid 	Householder, 2 parents 	Married couple, kid, parent-in-law 	Married couple, 6 kids 	Married couple, 2 grandkids 	Parent, 4 kids, friend or partner 

EU 1989



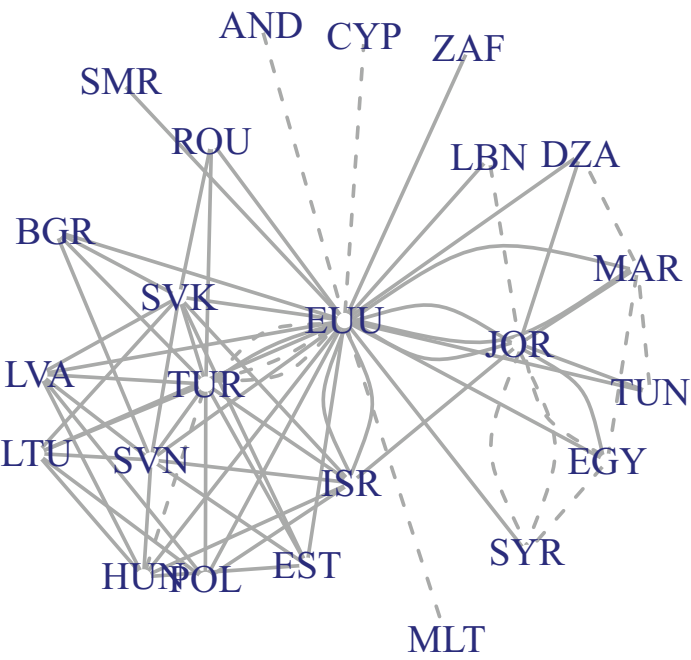
EU's Bilateral PTAs = 19
Transitivity = 0.06

EU 1994



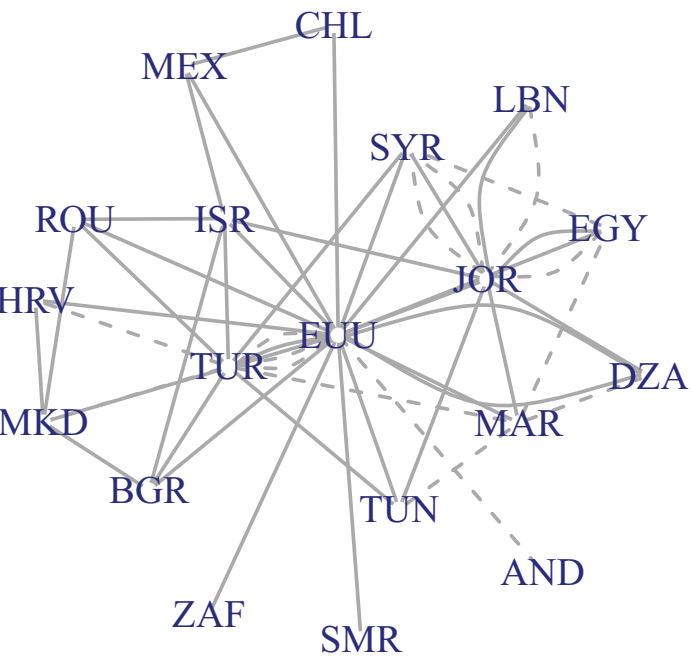
EU's Bilateral PTAs = 27
Transitivity = 0.15

EU 1999



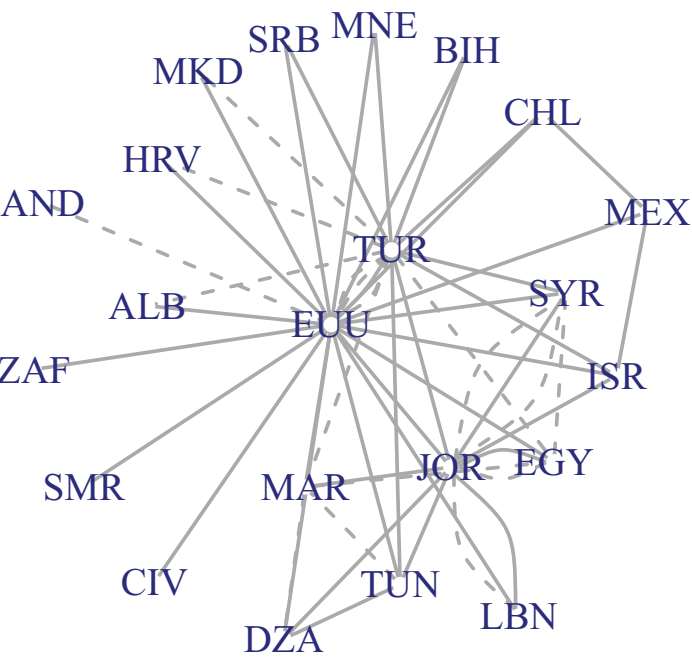
EU's Bilateral PTAs = 29
Transitivity = 0.41

EU 2004



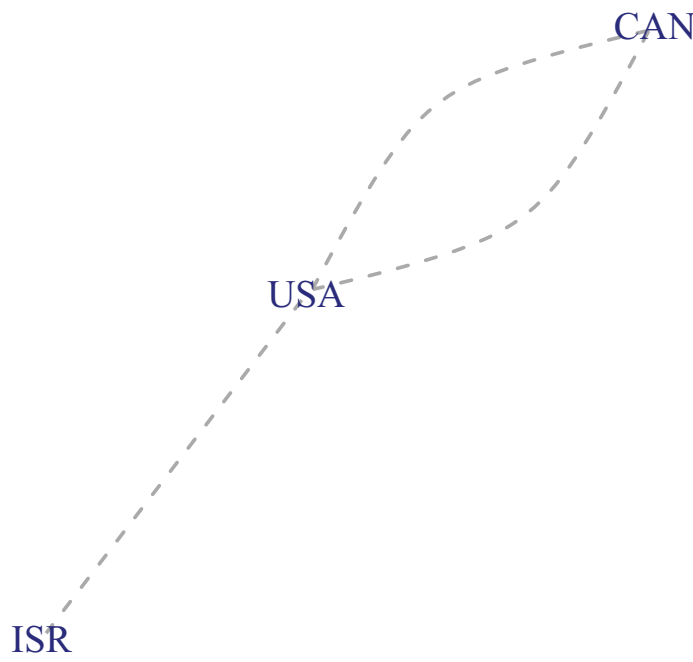
EU's Bilateral PTAs = 22
Transitivity = 0.36

EU 2009



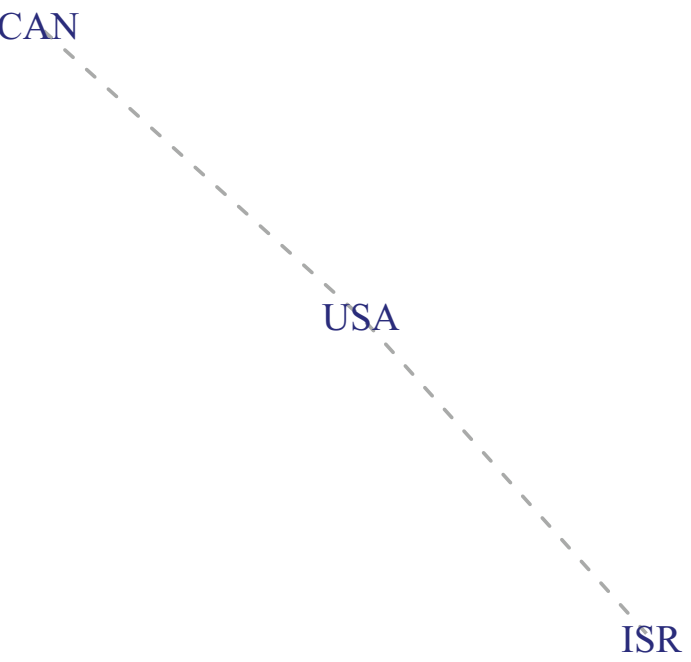
EU's Bilateral PTAs = 24
Transitivity = 0.31

USA 1989



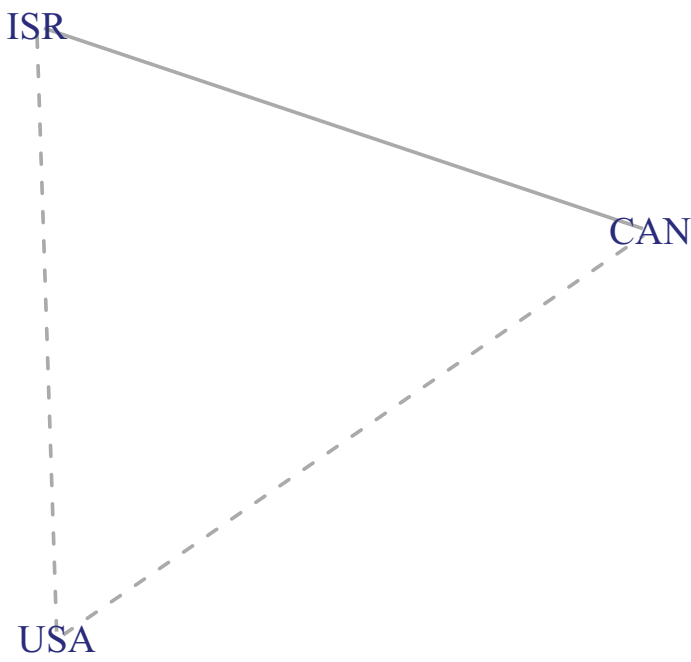
USA's Bilateral PTAs = 3
Transitivity = 0

USA 1994



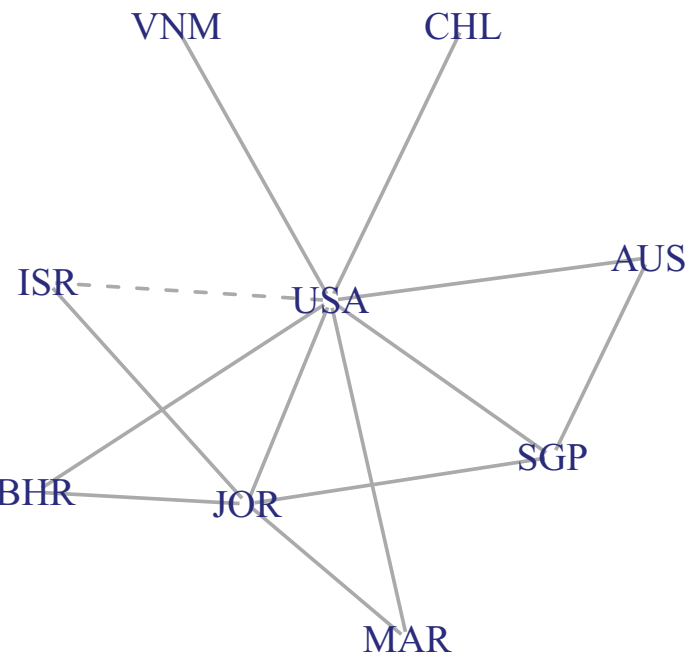
USA's Bilateral PTAs = 2
Transitivity = 0

USA 1999



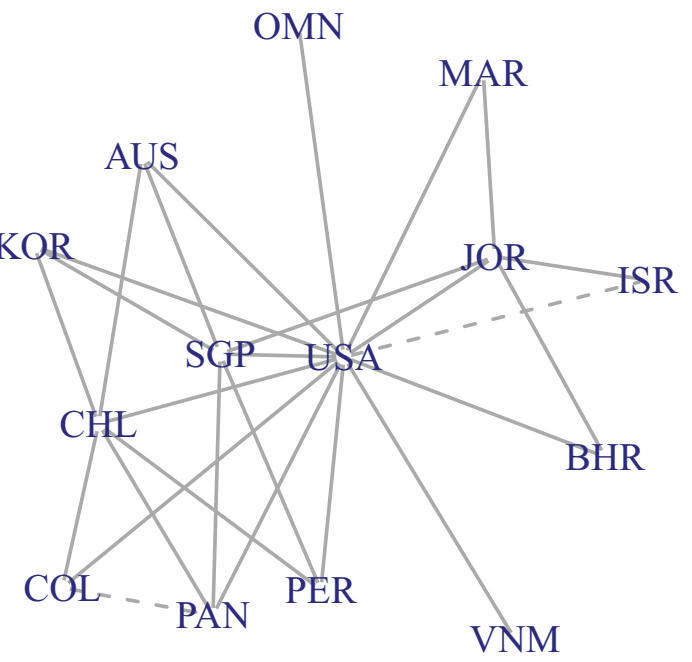
USA's Bilateral PTAs = 2
Transitivity = 1

USA 2004



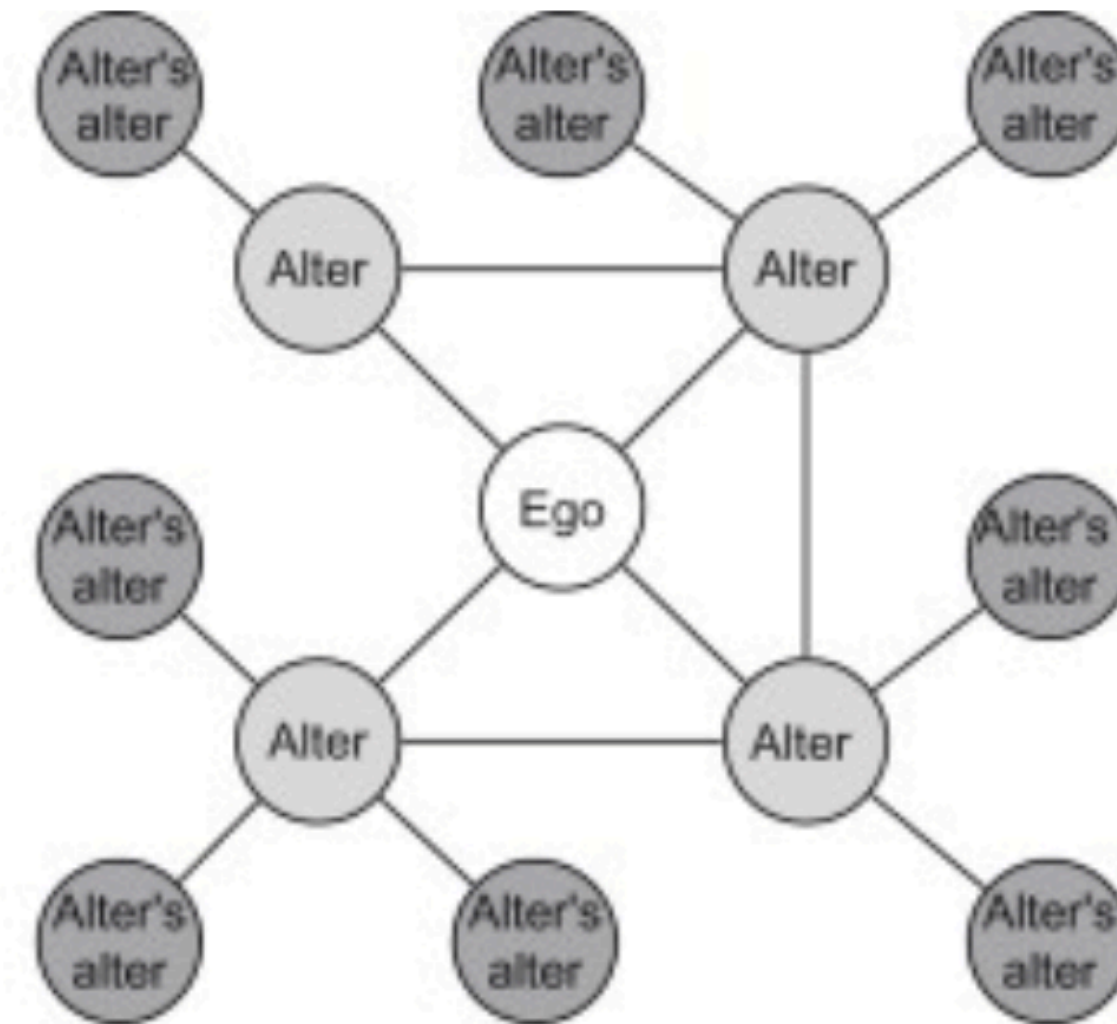
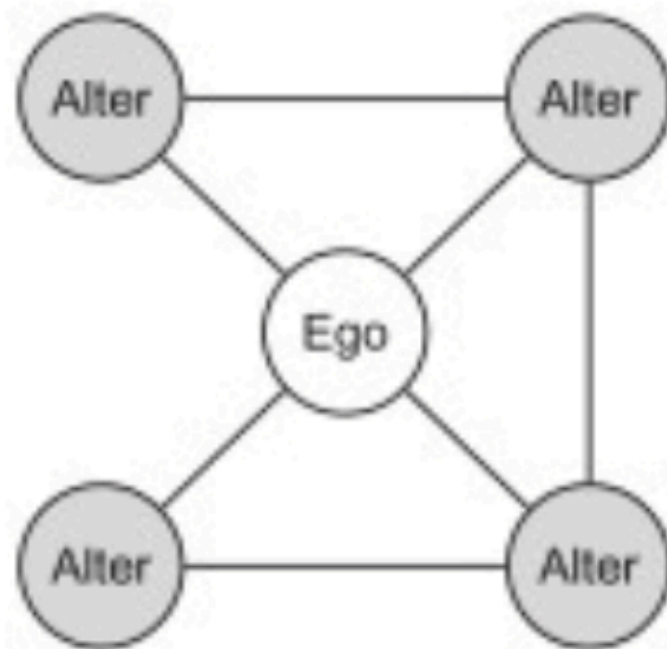
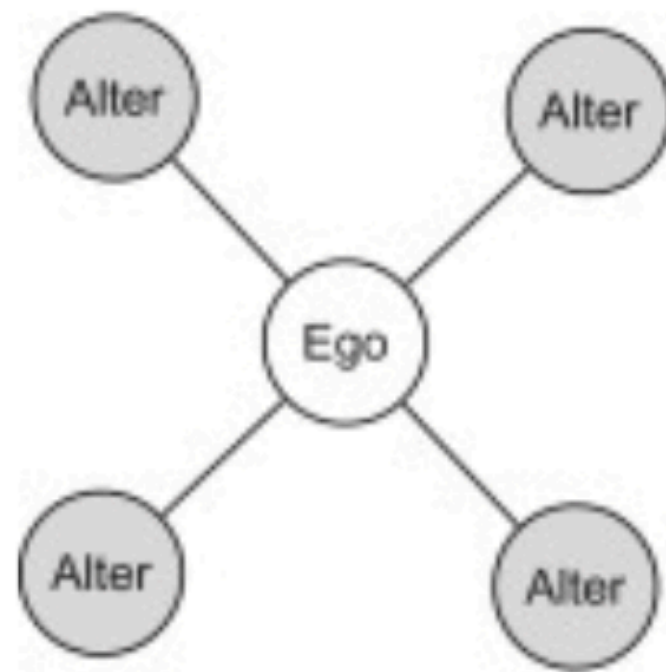
USA's Bilateral PTAs = 8
Transitivity = 0.33

USA 2009



USA's Bilateral PTAs = 13
Transitivity = 0.32

Ego networks



- Can analyse larger networks (sampling)
- Compatible with traditional methods (and designs)
- Can study intersecting social circles (through focal nodes)

- Demands traditional assumptions (exchangeability)
- Missing global context (betweenness)
- Recall often a significant challenge

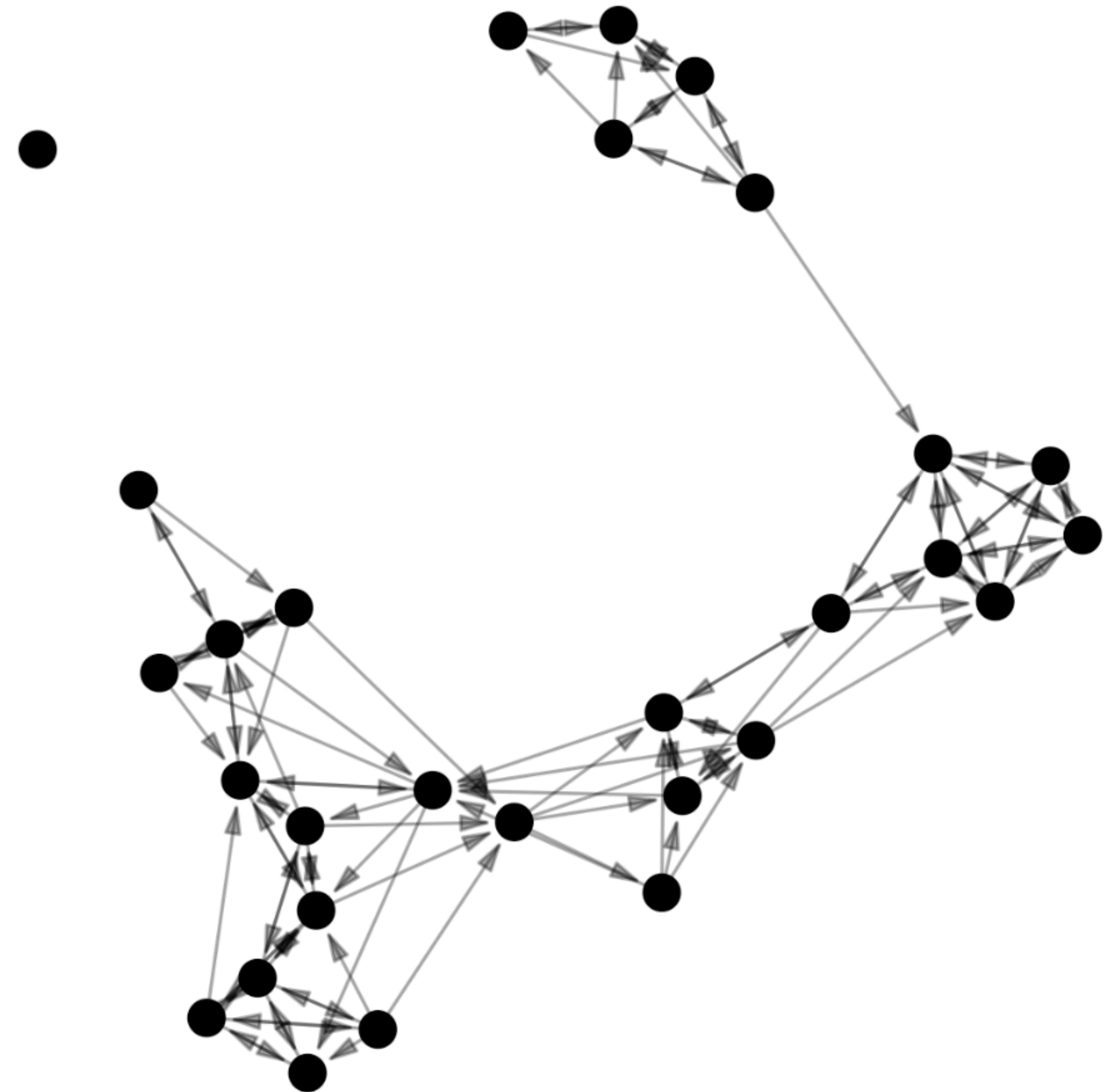
Check out `collect_ego()`

Important methodological issues

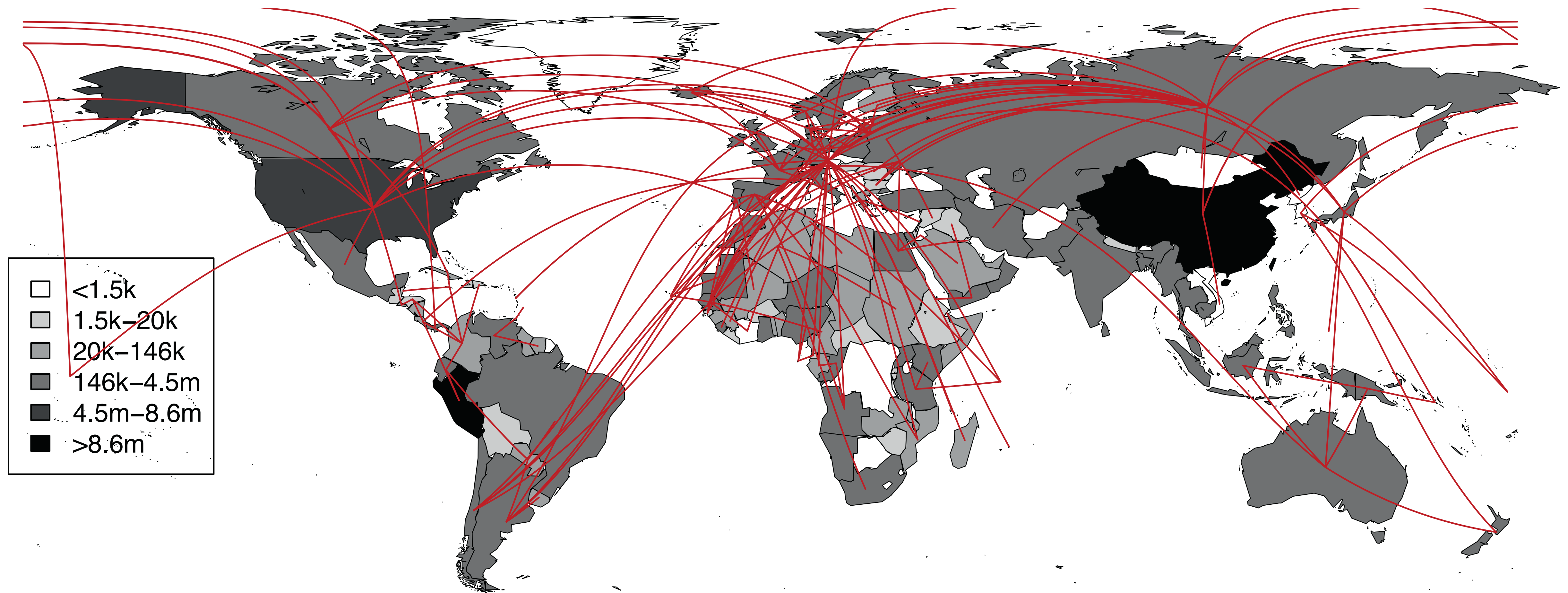
- While name generators commonly used method, some important methodological issues (e.g. Adams and Moody 2007, Campbell and Lee 1991, Ferligoj and Hlebec 1999, Marsden 1993, 2003, Matzat and Snijders 2010)
 - Recall often biased towards longer term interactions (Freeman, Romney, and Freeman 1987)
 - Recall often inaccurate when concerning third party relationships (McEvily 2014)
- E.g. McPherson et al. 2006 found personal networks of Americans had shrunk significantly between 1985 and 2004, but it turned out that this was an artefact and actually due to:
 - Interviewer fatigue and other interviewer effects (e.g. uneven prompting, Bearman and Parigi 2004)
 - Respondent fatigue (e.g. when later in the survey, Paik and Sanchagrin 2013)
 - Respondent learning (e.g. panel conditioning, Warren and Halpern-Manners 2012)

Whole networks

- Respondents presented with a **roster** (complete list of individuals in a population of interest; Marsden 1990)
- Reduces recall bias stemming from unreliably recalling interaction partners' names
- Reduces perception bias because reports cross-checked, or not even requested (though see Boda et al 2020)
- Only possible for small networks or networks where relational data publicly recorded
- Large rosters can still become unreliable (Pustejovsky & Spillane 2009)



Also, archives...

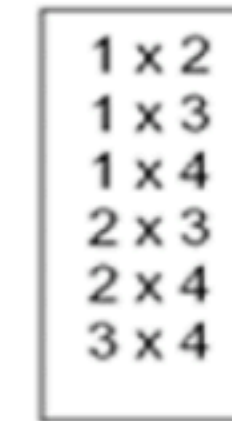


Cognitive social structure (CSS)

- Asks respondents to report on the structure of relations of *others* in the network from their point of view
- Then similarities and discrepancies between the network as reported by individuals in the network is analysed
- Or impacts of perceived social structure on outcome is considered, e.g. perceived influence/popularity

See Krackhardt 1987; Knoke 1998; Knoke et al. 2019

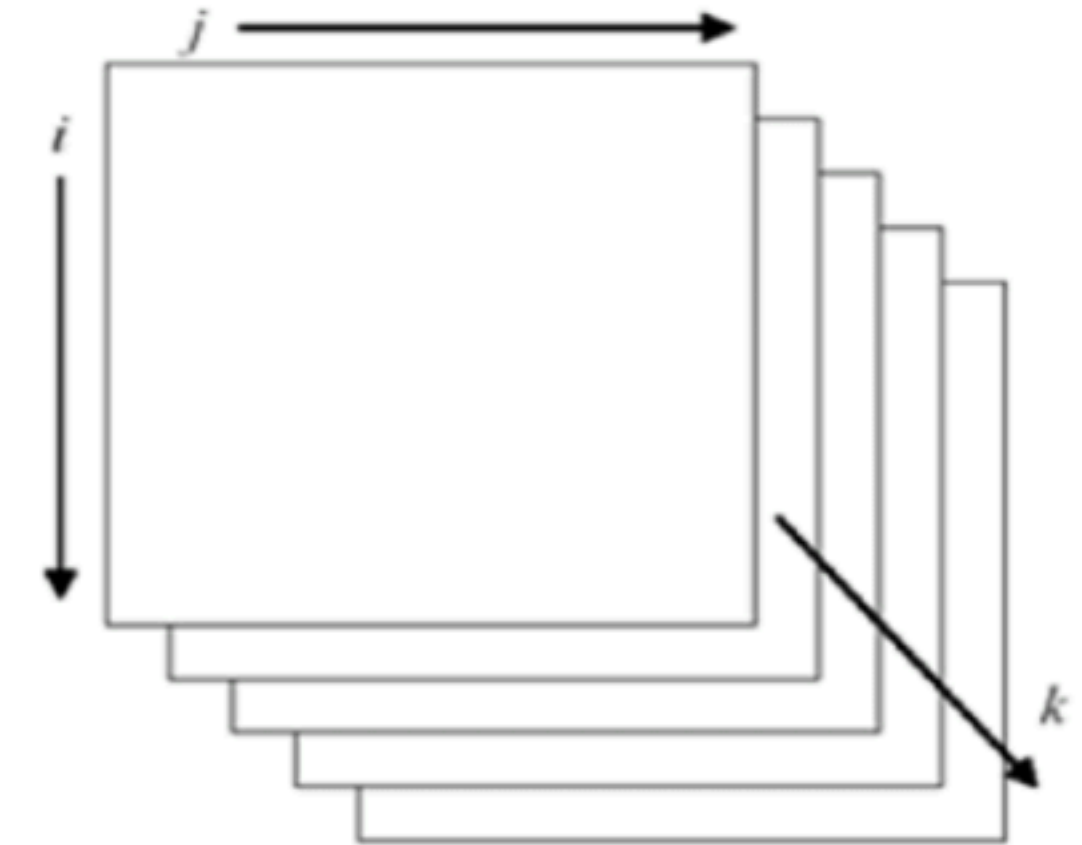
1. Participants are asked about the support for H1N1 vaccination shared by each possible pair of individuals in the group:



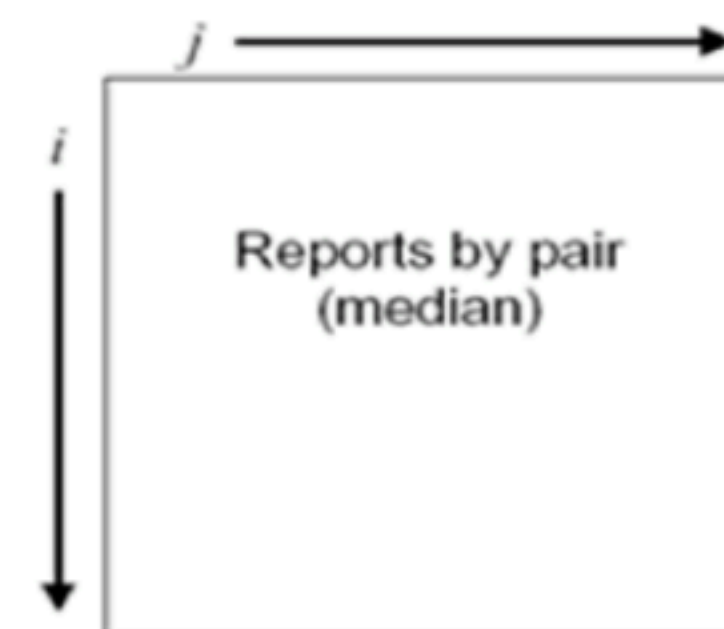
2. Which is translated into a 2-dimensional ($i \times j$) matrix form called a slice:



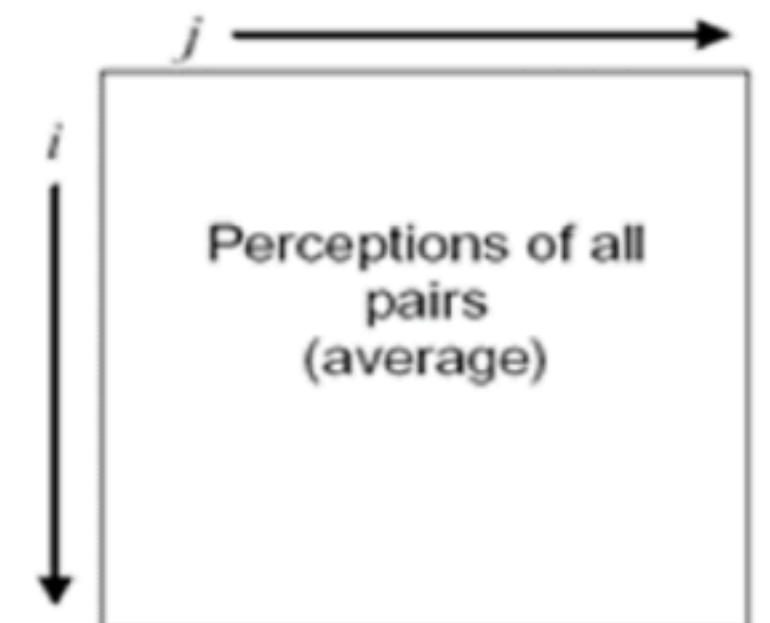
3. Together all of the slices from each individual participant create the 3-dimensional ($i \times j \times k$) cognitive social structure:



4. From which are extracted a locally aggregated structure ($k = i \cap j$) that is what is reported by the individual pairs about their interactions:



5. And a consensus structure that shows what the average perception of the interaction between each pair is:



Ego's Perception of Relationship Frequency and Valence

P01

Studies in Social Relationships and Health

Y10 Personal Network Structure

Case-ID	Row #	Name	F	V	F	V	F	V	F	V	F	V	F	V	F	V	F	V
	1																	
	2																	
	3																	
	4																	
	5																	
	6																	
	7																	
	8																	
	9																	
	10																	

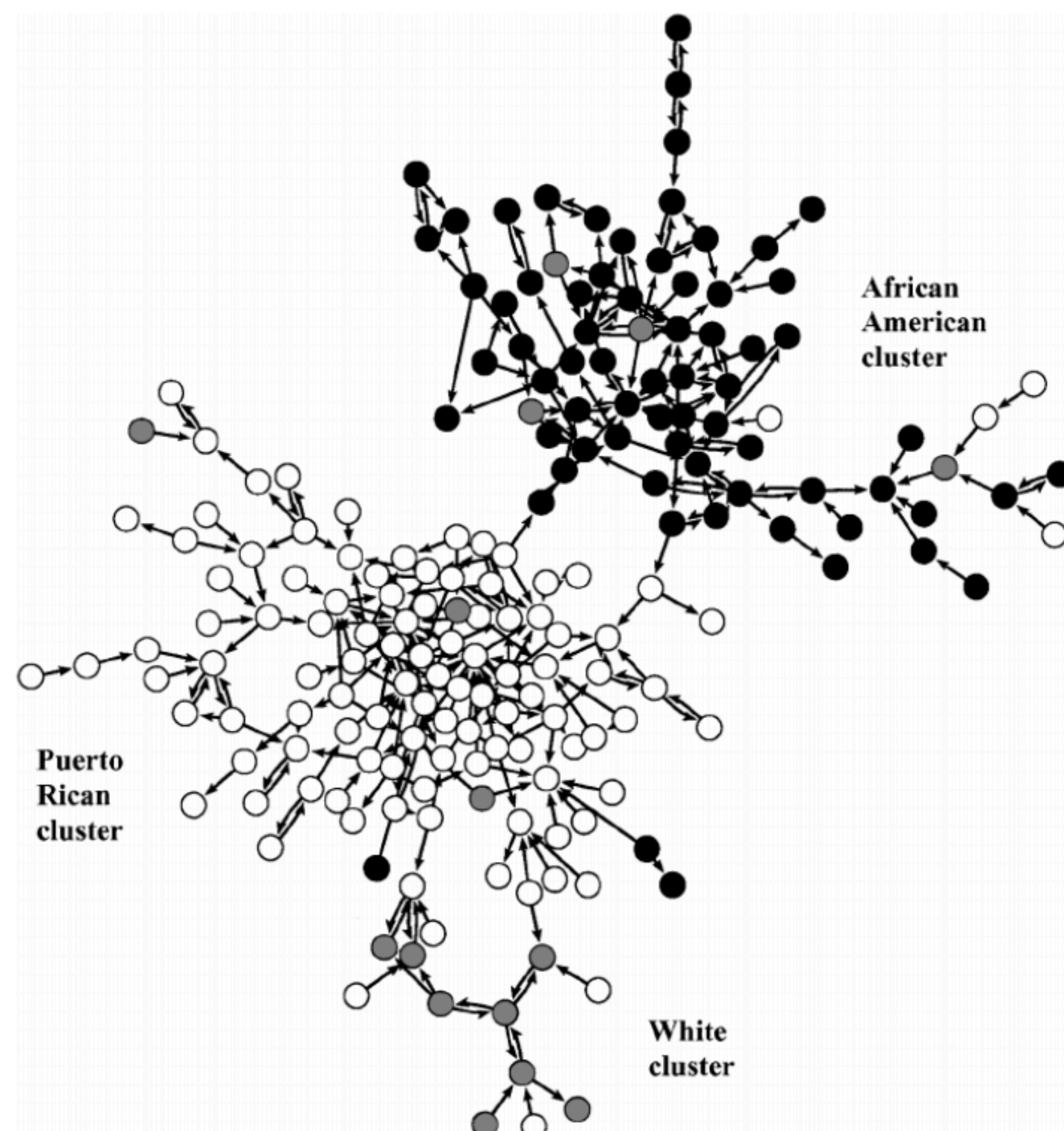
Frequency of Contact:

1) Every day
2) Several times a week
3) Once a week
4) Once every two weeks
5) Once a month
6) A couple of times a year
7) Once a year
8) Less than once a year
9) Have never spoken

Valence

-3) Dislike each other
-2)
-1)
0) Neither like or dislike
1)
2)
3) Like each other

Snowball sampling



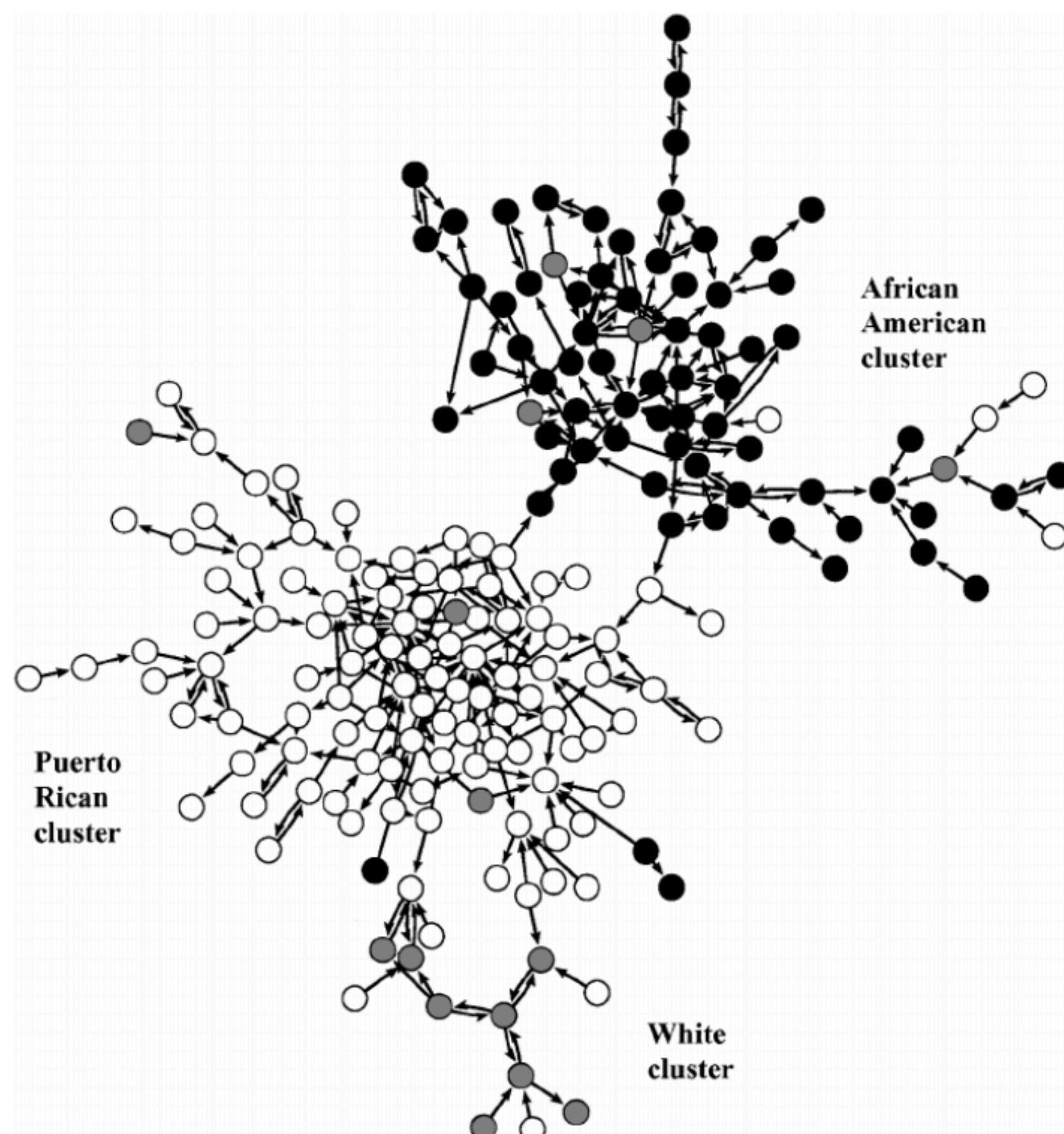
some **seed nodes** recommend others, who recommend more, usw; a non-probability sampling method that concentrates on most accessible part of the population

- cheap, simple, and cost-efficient
- easy way into hard-to-reach (small, covert) groups

But

Needle sharing among IV drug users in Hartford, CT

Snowball sampling



Needle sharing among IV drug users in Hartford, CT

some **seed nodes** recommend others, who recommend more, usw; a non-probability sampling method that concentrates on most accessible part of the population

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- easy way into hard-to-reach (small, covert) groups

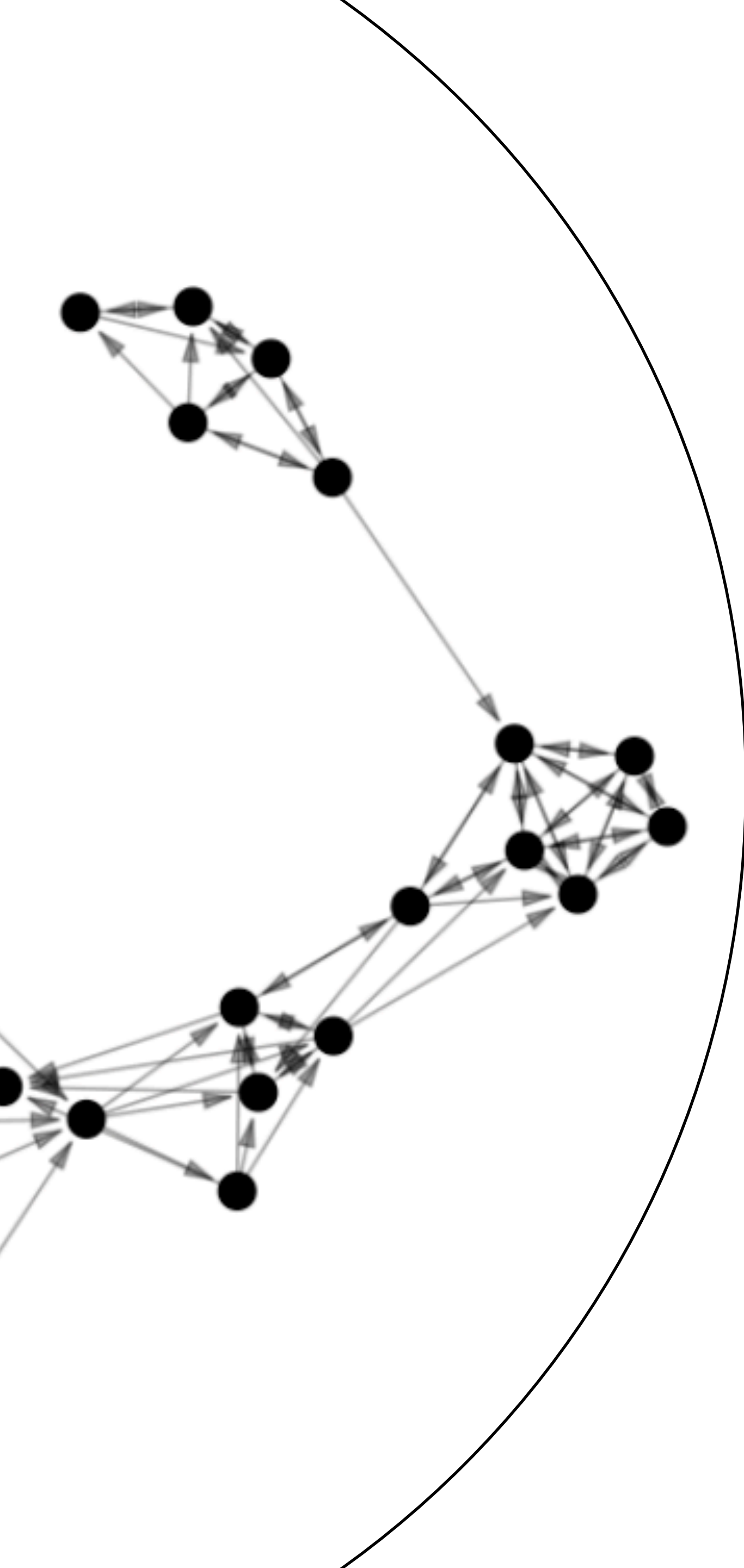
But

- oversamples most public and well connected (more often seeds and lie on more recruitment paths) and may miss isolated/weakly connected individuals/groups
- biases towards particular network structures (like high degree) and not representative

Respondent-driven sampling (RDS) aims to mitigate some of these concerns by weighting the sample to compensate for non-random recruitment patterns

Boundaries

- **Realist approach** (the nodes' perspective):
 - i.e. analyst discovers the network's boundaries from the nodes
- **Reputational approach** (i.e. relational):
 - e.g. all 'relations' connected to seeds
- **Participatory approach** (i.e. event-based):
 - e.g. all 'elites' that show up at relevant meetings
- **Nominalist approach** (the analyst's perspective):
 - i.e. analyst imposes network boundaries from theory
 - **Positional approach** (i.e. formal offices):
 - e.g. all 'senators' holding office in the senate
 - **Attributional approach** (i.e. characteristics):
 - e.g. all 'populist politicians' based on programmatic positioning



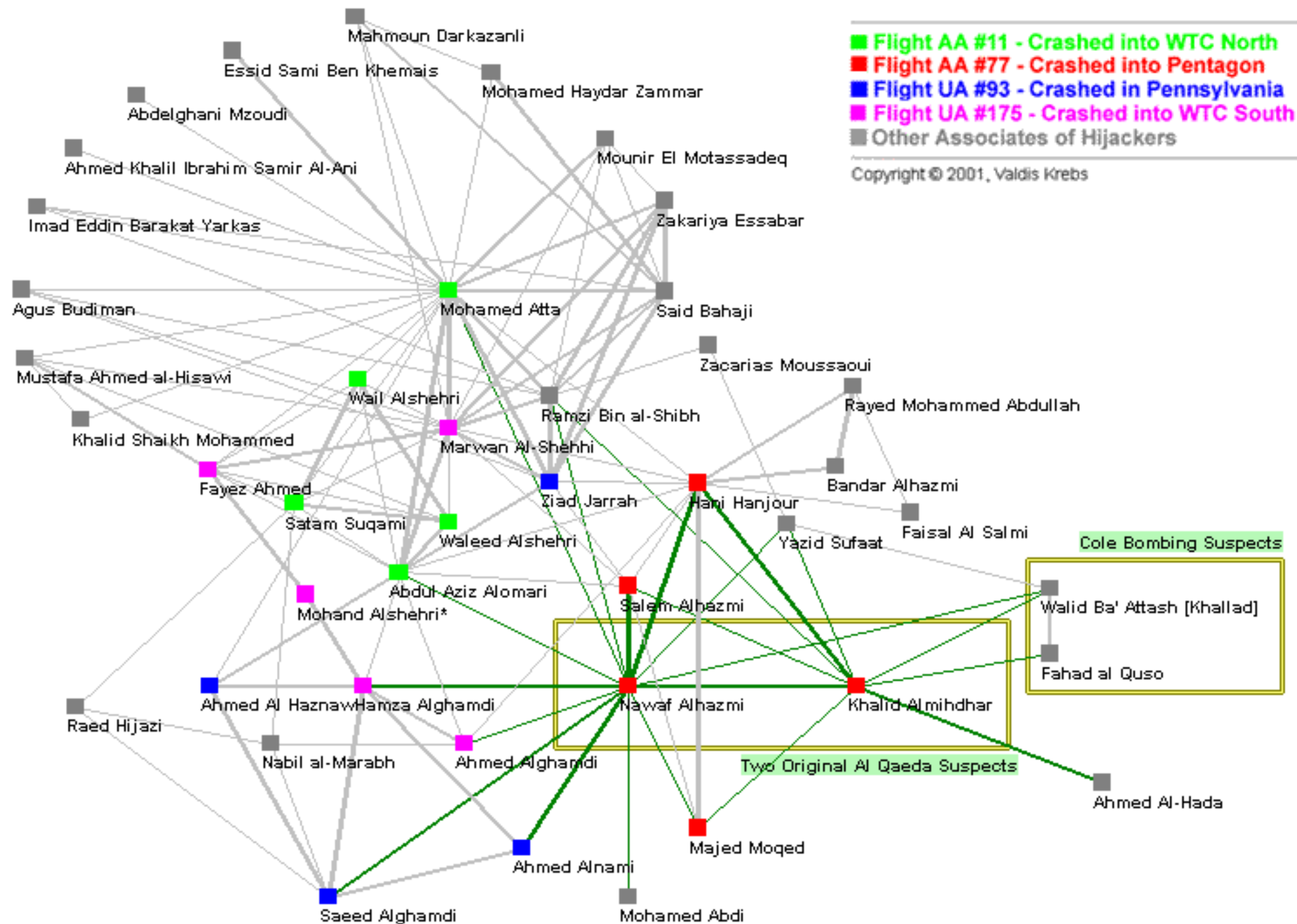
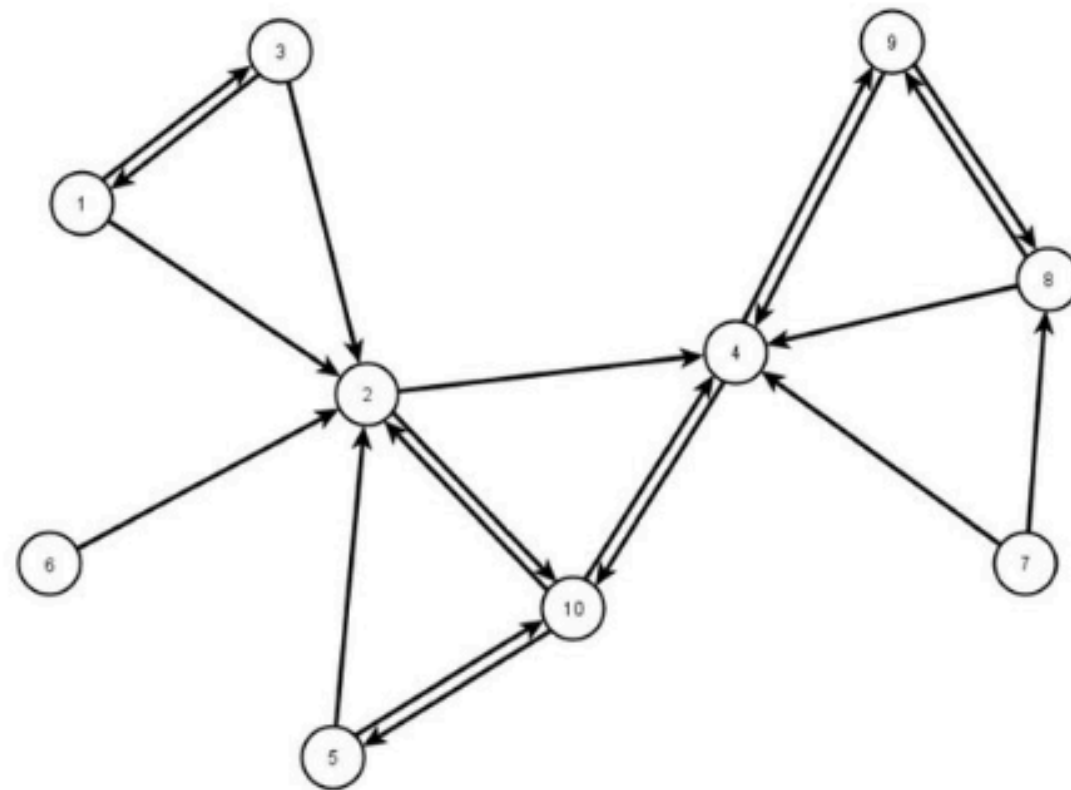


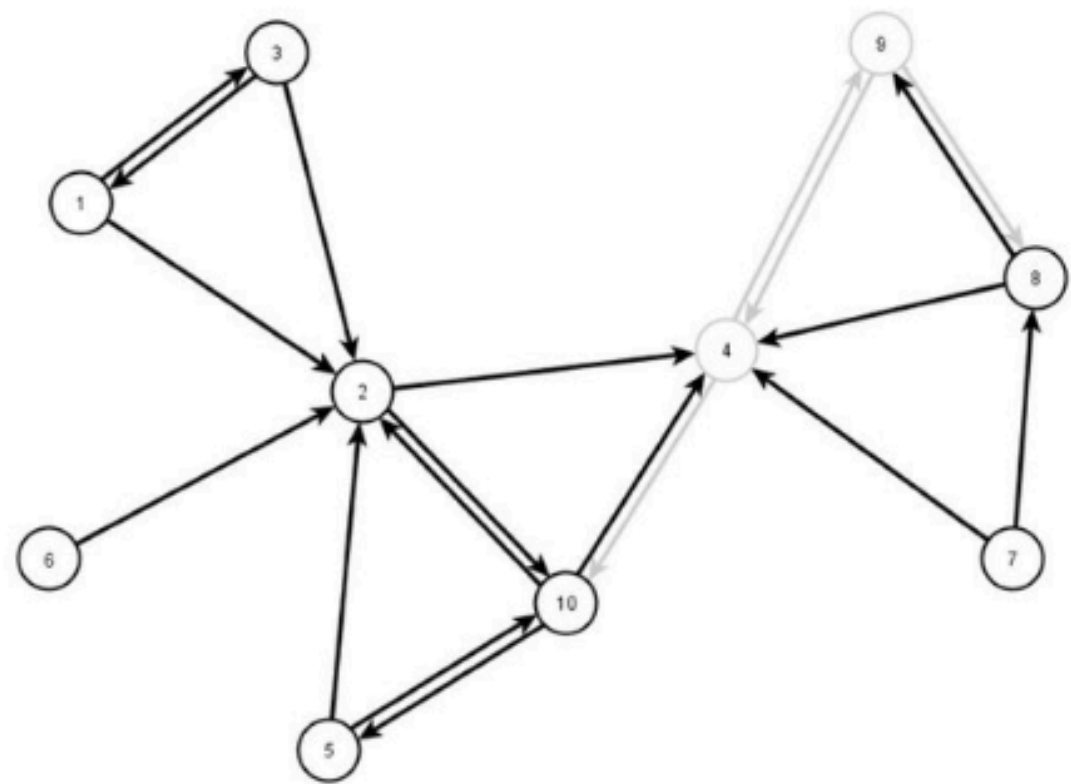
Figure 3 - All Nodes within 2 steps / degrees of original suspects

Missing data

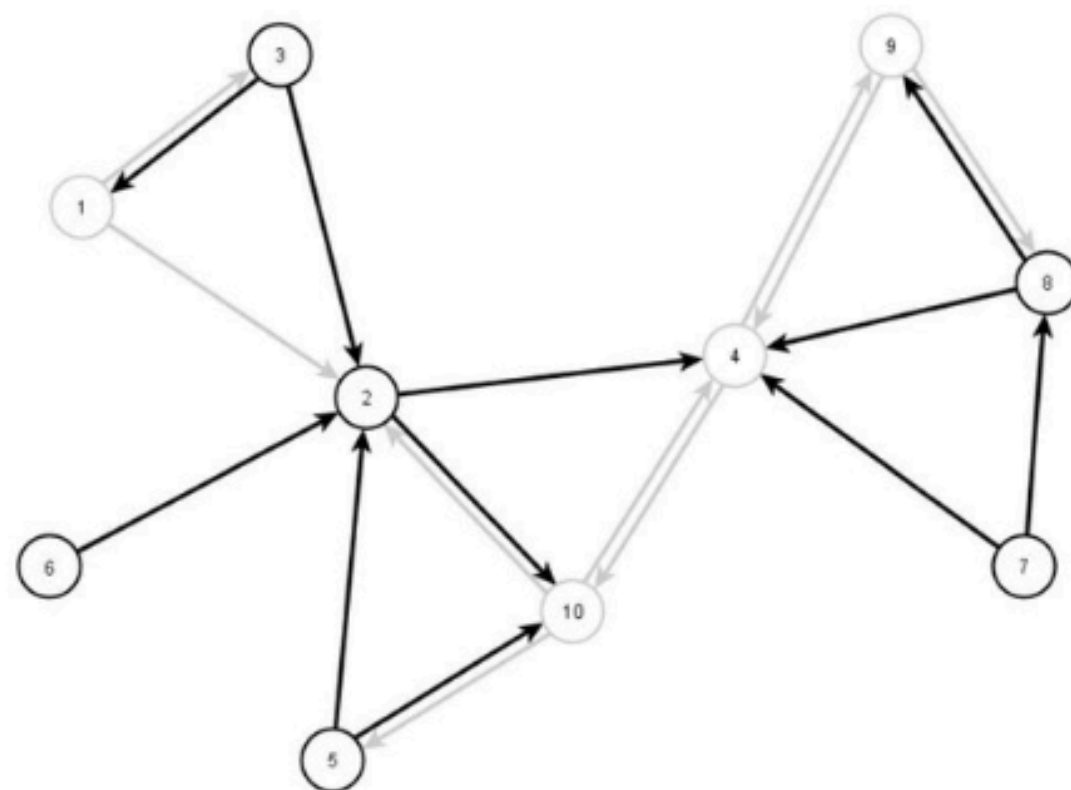
100% participation



80% participation



60% participation



- Unfortunately, even a small fraction of missing observations can be problematic
- A single non-response in a large survey is 1 missing observation
- But a single non-response in a network survey is $n - 1$ missing observations
- It can significantly bias network structure
 - What do you do with missing targets?
 - Can skip how communities connected at the margin
 - Moreover, missing often not at random...

Ethical concerns

- Consider study ethics from initial research design
 - IRB/MPT reviews
 - Report in publications
- Study participation needs
 - Voluntary, informed consent
 - Minimal intrusiveness
 - Appropriate protections of privacy
- One of the main advantages of networks is also a source of greatest ethical concern...
 - *What is it?*





Lesson #3

Data collection affects
data analysis

GRAND

Network Facts

Zachary Karate Club

Nodes

People	34
--------	----

Edges

Non-club interactions (<i>Undirected, Unweighted</i>)	78
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Source	Observation
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Location	American University
----------	---------------------

Time	1970 - 1972
------	-------------

Details in	10.1086/jar.33.4.3629752
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Transformations

None

Demanding data

- Networks are demanding of data
 - Yet true of all attempts at providing persuasive evidence...
- Where data comes from crucial because:
 - how meaningful your descriptive or inferential conclusions depends on tie-data being *meaningful*
- **Most important lesson:** always ask what a tie *means*
 - i.e. how can we interpret structures relationally



Tie content

- *Actions*

- Talks to, gives advice to, gives support to
- Sells to, transfers to, trades with
- Sleeps with, fights with

- *Cognitive/affective*

- Recognises, knows
- Likes, loves, despises
- Trusts/distrusts

- *Role-based*

- Kinship: brother of, daughter of
- Social: friend of, competitor of, representative of, rival of
- Organisational: boss of, teacher of, member of, affiliate of



Tie content

- *Actions*

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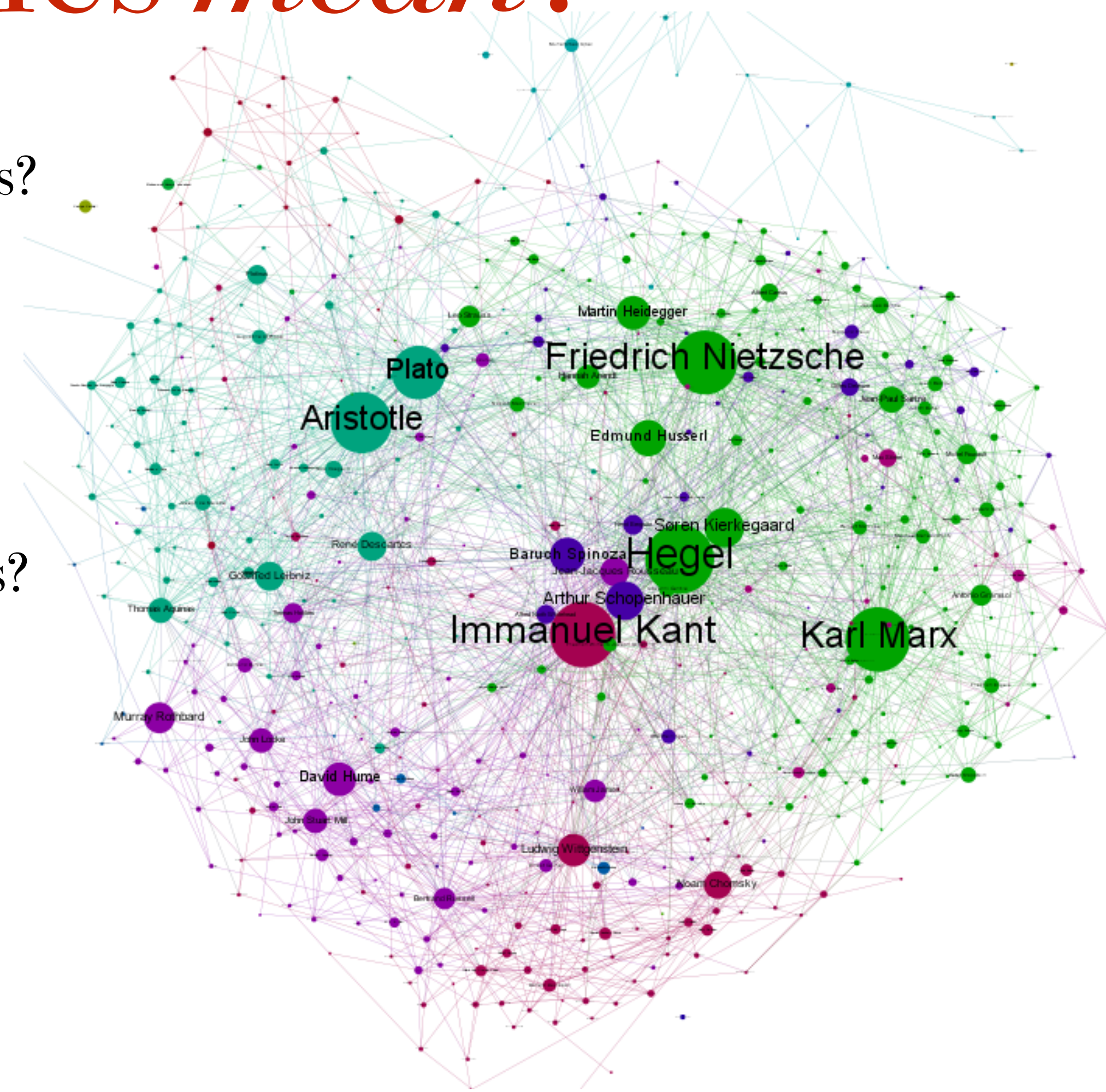
- Kinship: brother of, daughter of
- Social: friend of, competitor of, representative of, rival of
- Organisational: boss of, teacher of, member of, affiliate of



Ask: what is *at stake*?
What is the tie *doing* or *for*?

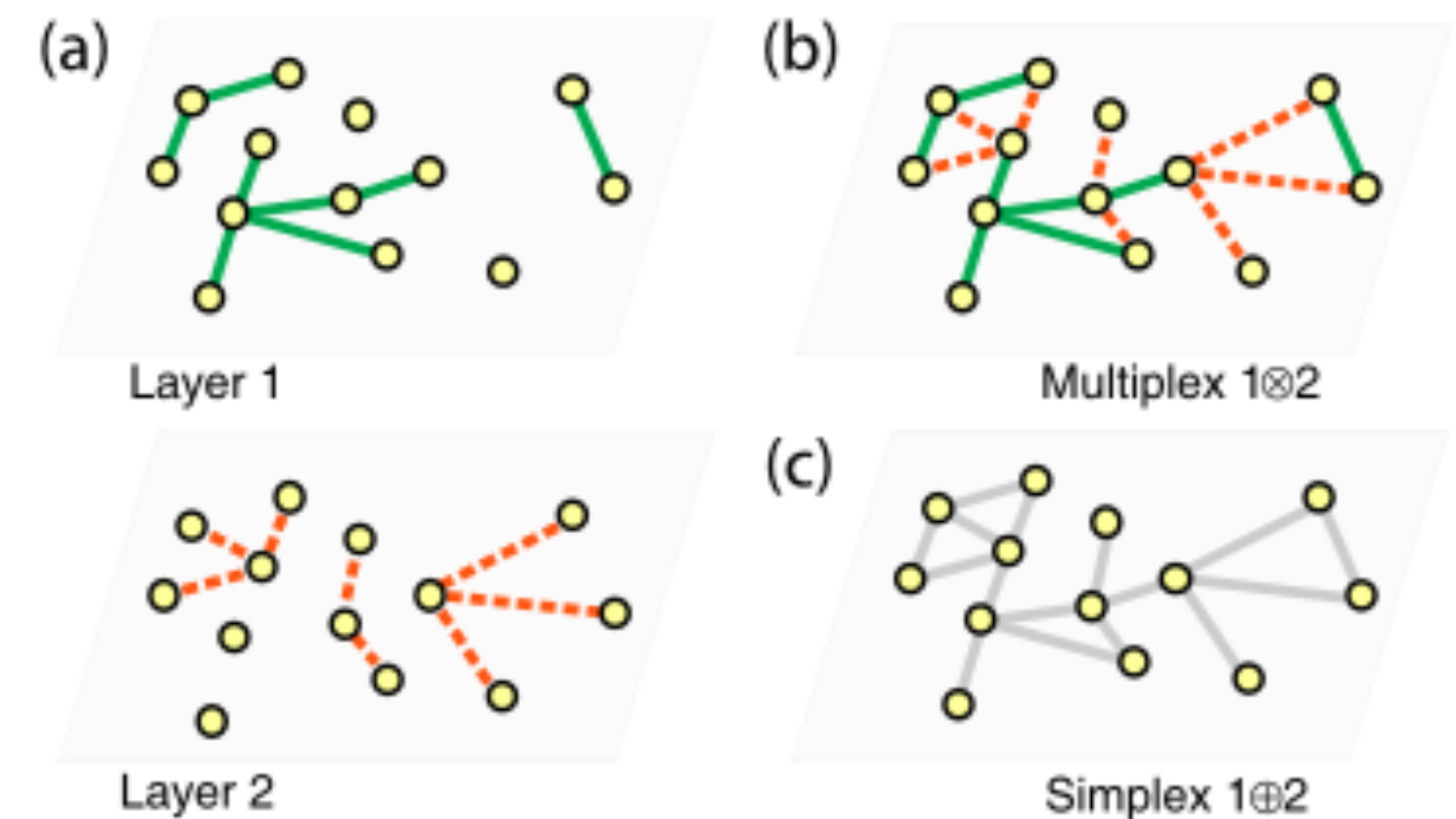
What do you think these ties *mean*?

- A network of philosophers on Wikipedia influenced by other philosophers?
- A network of kinship in a royal family over time?
- A network of citations between articles on a topic for the past five years?
- A network of co-citations between articles on a topic for the past five years?
- A network of hyperlinks between NGO websites?
- A network of keyword co-occurrences in texts?
- A network of retweets (sic) between accounts/posts?
- A network of who knows who between class members?

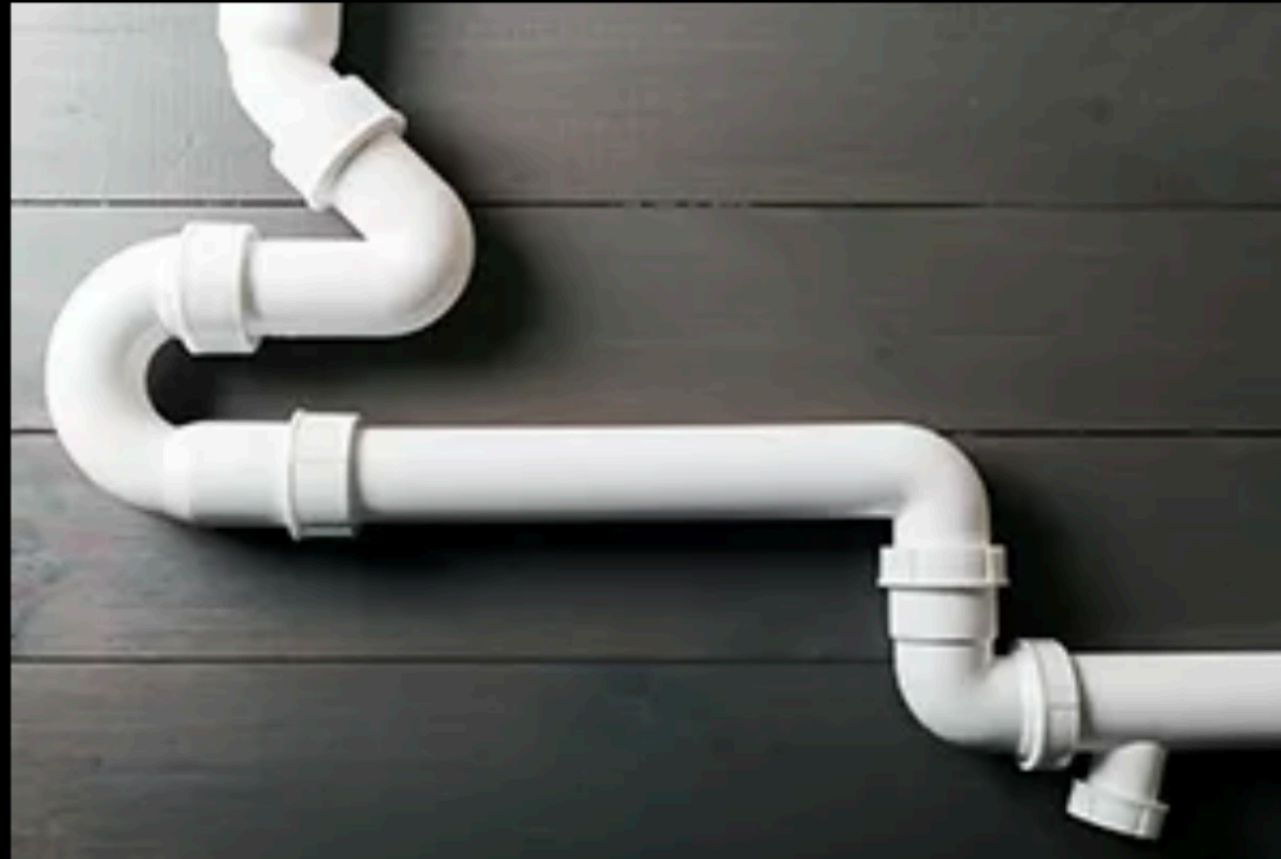


Meaning trade-offs

- Should a tie be defined as *any* of e.g. “friendship/advice/support/hanging out/messaging” or *only* one of these?
 - *any* may give you more observations
 - *only* gives you more interpretability
 - *and/or* makes it a multiplex network
-
- Domain experts, what data is accessible? What relationships make sense? Is there a satisfactory proxy available?

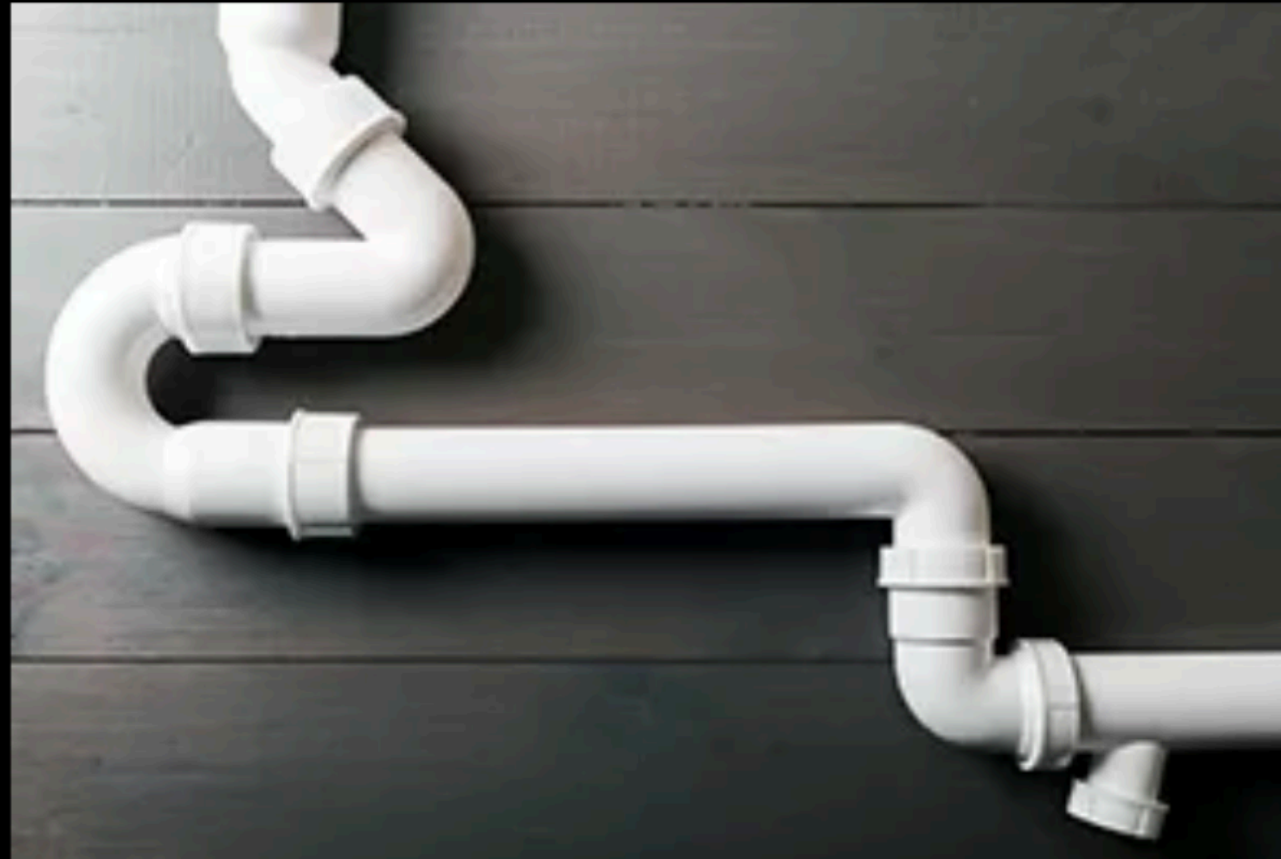


Pipes



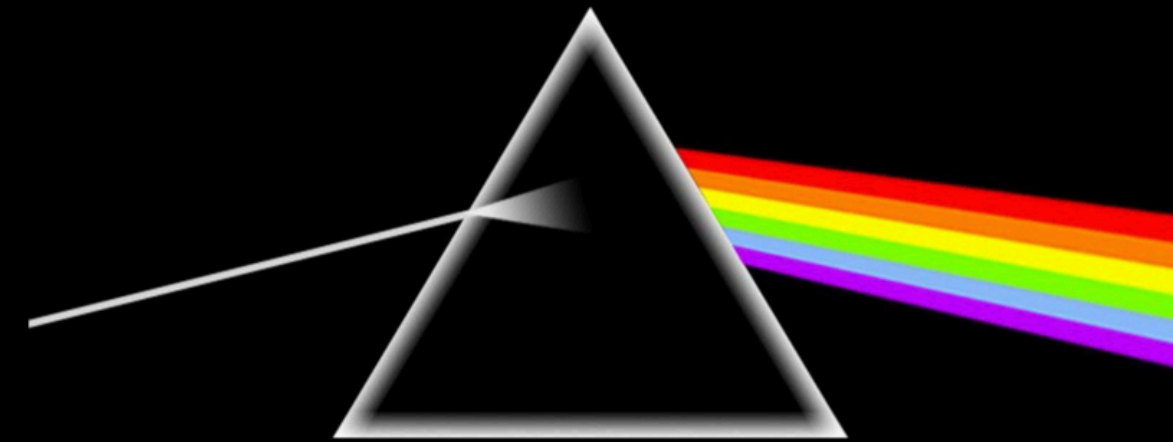
- Ties often seen as ‘plumbing’ through which ‘stuff’ flows
- Stuff can be ideas, capital, etc
- Micro Qs about position, inequality, etc
- Macro Qs about intervention, resilience, etc

Pipes



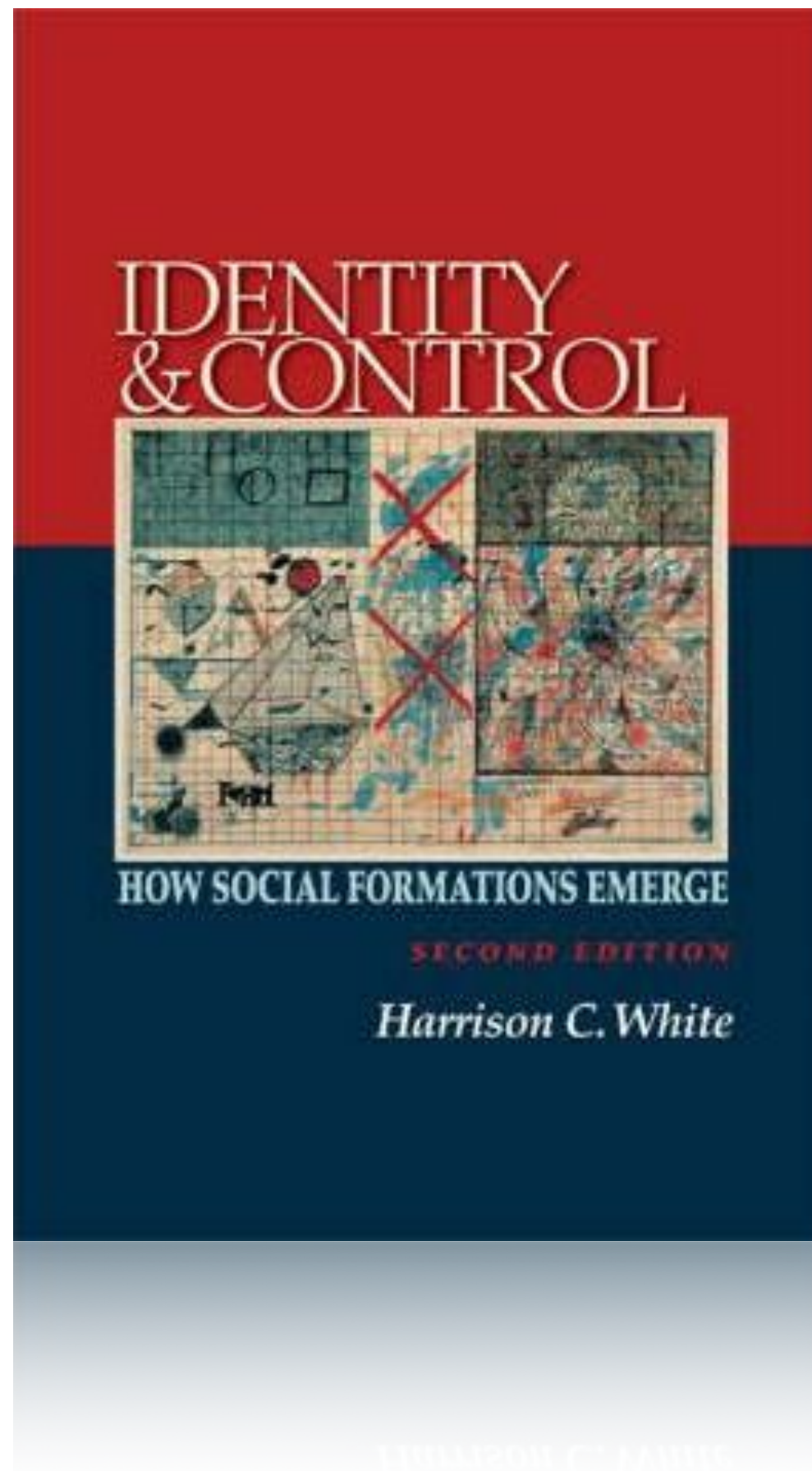
- Ties often seen as ‘plumbing’ through which ‘stuff’ flows
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- Macro Qs about intervention, resilience, etc

Prisms



- Ties also serve as informational ‘cues’
- Cues can be to identity, preferences, etc
- Micro Qs about perception, preferences
- Macro Qs about culture, emergence

Networks and Culture



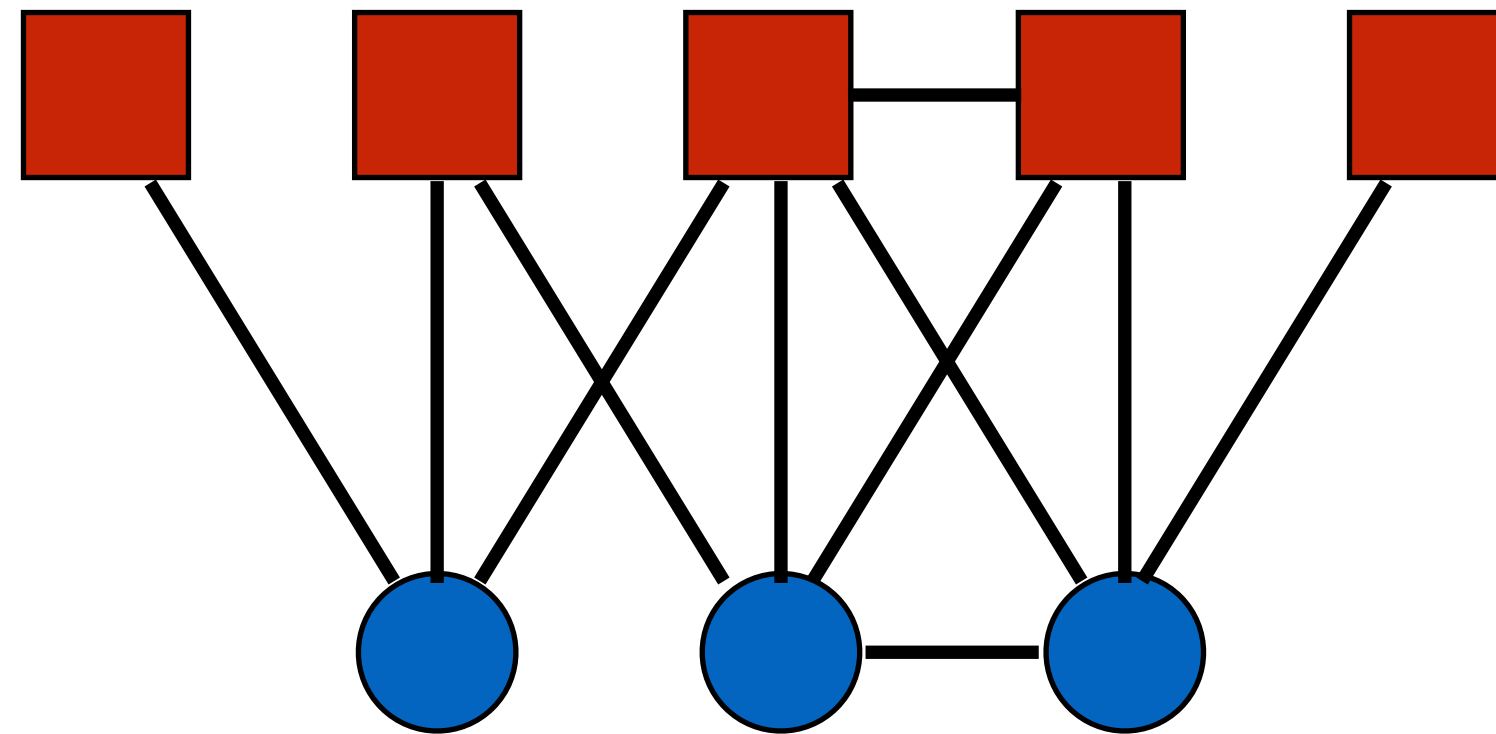
- *structure* long linked with *culture* (e.g. Berger and Luckmann, Bourdieu, Douglas)
- 1970s “breakthrough” established social networks as method of *structural* analysis distinct from *cultural* analysis and Parsonian normative theory (Blau, White et al 1976)
- today tentative reconciliation in different directions:
 - *ties affect culture* (Erickson, Krackhardt, Kilduff, Carley, DiMaggio)
 - *culture affects ties* (symbolic interactionism/cultural sociology of Emirbayer and Goodwin, Lizardo, etc)
 - a *duality* understanding (Breiger 1974, White 1992, Mische, Fuhse)

Duality

- Sociology of culture explores duality of persons and groups, persons and cultural elements, etc.
- *Dualism*: bifurcating to separate things into distinct categories
- *Duality*: recognition that thing has another, inseparable side to it
- I.e. we can understand a field of political action both from the perspective of politicians expressing beliefs but also beliefs affiliated to by politicians



Multilevel Meaning



Socio-semantic networks: Roth et al

Socio-ecological networks: Bodin et al

Socio-organizational networks: Hollway et al

Socio-political networks: Knoke et al

See `is_multilevel()` and `to_multilevel()`



Lesson #4

Meaning matters

Getting started



```
- migraph::run_tute("Data")
```